

RESTAURANT MANAGEMENT SYSTEM

Inventory, Stock-Taking, Menu Management and POS
Software Development Blueprint

MODULES	SCREENS	MARKET	PHASES
11 sections	10 + mobile	Kenya (KSh, M-Pesa)	4

Version 1.0 | 6 November 2026 | For the restaurant owner and the appointed software developer

Sample restaurant name, people, suppliers and figures are fictional and used for illustration.

Contents

1 Project overview . . . 3

2 System objectives . . . 4

3 Complete module descriptions . . . 5

4 Inventory calculation formulas . . . 7

5 Stock-taking workflows . . . 10

6 Makeline workflow . . . 12

7 Menu management . . . 13

8 POS and order processing . . . 14

9 Kitchen ticket workflow . . . 16

10 Reporting requirements . . . 17

11 User roles and permissions . . . 18

12 Database structure and technical requirements . . . 19

13 Important business rules . . . 24

14 UI mockups . . . 25

15 Sample data . . . 37

16 Development phases . . . 39

17 Testing requirements . . . 40

18 Developer acceptance checklist . . . 42

19 Future improvements . . . 43

A Appendix A - Step-by-step user workflows . . . 44

B Appendix B - Owner decisions register . . . 48

How this blueprint is organised

Sections 1-3 explain the business and the modules. Sections 4-13 define the rules, formulas, workflows and database. Section 14 shows the screens. Sections 15-19 give sample data, phases, tests, acceptance checklist and future plans. Appendix A has click-by-click workflows and Appendix B lists every decision the owner must approve.

1 Project overview

This document is the complete blueprint for a **Restaurant Inventory, Stock-Taking, Menu Management and Point-of-Sale (POS) System** for a restaurant in Kenya. It is written for two readers at the same time: the **owner**, who needs to confirm that the system matches how the restaurant really works, and the **software developer**, who needs enough detail to estimate, design and build it without guessing.

The restaurant buys raw ingredients from suppliers, prepares some items in the kitchen ahead of service (marinated chicken, sauces, dough), sells food and drinks to customers in the restaurant and through delivery platforms, and counts what is left at the end of each day. The system connects all of these activities so that the owner can see, at any time, what was bought, what was used, what was wasted, what was sold and how much VAT was collected.

1.1 What the system does, in one paragraph

Employees record **stock received** from suppliers and **makeline production** from the kitchen. Each day the manager enters the **physical closing stock**; the system works out how much of each product was used and what that stock was worth. Cashiers take customer orders on a fast **POS** screen, the kitchen sees them on **kitchen tickets**, and customers receive a **receipt**. Every transaction is stored permanently and can be searched. The owner sees a **dashboard** and **reports** in KSh, exportable to PDF and Excel.

1.2 Scope and market context (Kenya)

- Currency is Kenya Shillings (KSh). Amounts are stored to 2 decimal places and shown with thousands separators (KSh 4,750.00).
- Payment methods must include **cash, M-Pesa, card and bank transfer**, plus configurable extras. M-Pesa payments record the transaction code (e.g. SJK4H2L9QX).
- Delivery platforms in scope: **Glovo, Bolt Food, Uber Eats, SpeedyDrop, NAI Deliver, Yum Deliveries**. The administrator can add or deactivate platforms.
- VAT must be configurable. The 16% shown in the samples is a **sample value only**; the rate, and whether prices include VAT, are settings, never fixed in code.
- Customer phone numbers are personal data. Collect them only when needed (delivery), restrict who can see them, and do not export them in routine reports (Kenya Data Protection Act, 2019).
- Many Kenyan businesses must issue tax-compliant receipts through KRA's eTIMS system. This blueprint does **not** assume how the restaurant is registered; see the owner decision below.

OWNER DECISION REQUIRED

Decision: Is the restaurant required to issue KRA eTIMS-compliant invoices or receipts from this POS?

Recommended default: Build the receipt layout and the order data so a KRA eTIMS integration can be added later (Phase 4). In the meantime, record the restaurant's KRA PIN on receipts. The owner must confirm the requirement with the accountant before launch because it can change which receipt format is legal.

Owner to approve or change before development of this feature starts.

1.3 How to read this document

- Sections 2 to 13 explain *what* the system must do, with formulas, rules and workflows.
- Section 14 contains realistic **screen mockups** with sample data. Sections 15 to 19 give sample data, build phases, tests and the acceptance checklist.
- Appendix A gives click-by-click workflows (A to J). Appendix B lists every **decision the owner must approve**, each with a recommended default.
- Boxes marked **OWNER DECISION REQUIRED** appear where information was missing. The developer should build the recommended default unless the owner says otherwise.

2 System objectives

#	Objective	How it is measured
1	**Simple to operate.** A new employee can record a delivery or take an order after a short demonstration.	Receive a delivery in under 2 minutes. Take a 4-item order in under 45 seconds. Close stock for 11 products in under 10 minutes.
2	**Accurate.** Every stock figure and every shilling can be traced to a record and a person.	Stock and sales calculations reproduce the worked examples in this document to the cent.
3	**Fast during service.** Nothing slows the cashier or kitchen during busy periods.	POS screens respond in under 1 second on a mid-range tablet over normal restaurant Wi-Fi.
4	**Honest about stock.** The owner can see where stock goes: sold/used, wasted, adjusted.	Wastage is reported separately from normal use. Adjustments always show a reason.
5	**Protects history.** Nothing important is overwritten or deleted.	Corrections create new records. Closings, orders and purchases are permanent.
6	**Grows without rebuilding.** Recipes, supplier analytics and delivery integrations can be added later.	Database already has the hooks (recipe table, separate makeline product table) listed in Section 12.
7	**Suited to Kenya.** KSh, VAT, M-Pesa, local delivery platforms.	All of these are configurable settings.

Design principle

When there is a choice between a clever feature and a simple one, build the simple one. Every screen should have one obvious main action (Save receipt, Submit closing, Complete order). If a feature cannot be explained to a cashier in two sentences, it belongs in a later phase.

3 Complete module descriptions

The system has 11 sections, shown in the sidebar menu. The diagram below shows how information moves between them. Read it left to right: purchases feed stock, stock feeds closing, closing feeds history and reports; menu items feed the POS, orders feed the kitchen and sales history.

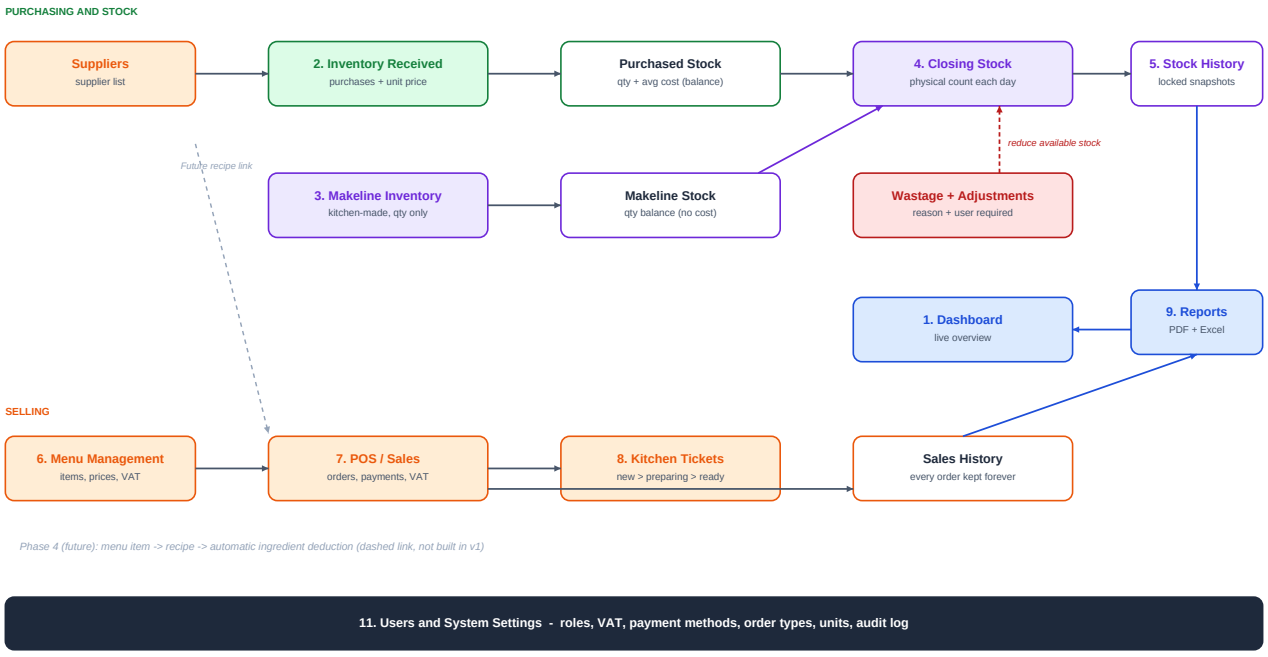


Figure 3.1 - How information moves between the 11 sections. Numbers match the sidebar order.

Section	What it does	Data it uses
1. Dashboard	Quick overview for owner/manager: today's sales, orders, average order value, sales by order type and payment method, best sellers, stock received, stock used value, wastage, low-stock and expiry alerts, quick-action buttons.	Reads from: orders, payments, purchases, closings, wastage, products. Writes nothing.
2. Inventory Received	Records every supplier delivery: product, quantity, unit, total price, supplier, invoice number, date, expiry date, who entered it. Unit price is calculated automatically. Supports multiple deliveries of the same product, supplier returns and damaged deliveries.	Writes: purchases, purchase_items. Increases purchased stock. Feeds Closing Stock (Received column).
3. Makeline Inventory	Records items prepared in the kitchen (prepared chicken, sauces, dough). Quantity, unit, date/time, person. Shown separately from purchased stock. Wastage and adjustments allowed. No cost value in version 1.	Writes: makeline_production. Feeds Closing Stock (Produced column).
4. Stock Taking / Closing Stock	The manager enters the physical quantity left for every purchased and makeline product. The system calculates available, used and value used, then saves a locked snapshot and sets the next opening stock.	Reads: opening, receipts, adjustments, wastage. Writes: stock_closing_sessions, stock_closing_items.
5. Stock History	Browse any past closing by date, compare two days, search by product, see received/used/wasted/adjusted, export to Excel and PDF.	Reads: closing snapshots only (never recalculated).
6. Menu Management	Create and maintain sellable items: name, price, category, description, image, availability, VAT setting, item code. Price-change history is kept. Items appear in the POS automatically.	Writes: menu_categories, menu_items, menu_item_price_history.
7. POS / Sales	Fast order entry: choose order type, add items, adjust quantities, apply authorised discounts, VAT calculated automatically, take payment, print receipt, send to kitchen.	Writes: orders, order_items, payments, kitchen_tickets.
8. Kitchen Order Tickets	Shows new orders to the kitchen, prints thermal tickets, tracks status New > Preparing > Ready > Served/Collected, or Cancelled.	Reads/updates: kitchen_tickets, orders.status.
9. Reports	Sales, VAT, stock received, stock used, wastage, closing history, employee sales; filter by date; export PDF and Excel.	Reads from every other section.

Section	What it does	Data it uses
10. Suppliers	Supplier list with contact details, plus purchase history per supplier. Used by Inventory Received.	Writes: suppliers.
11. Users and System Settings	Users, roles and permissions, VAT, payment methods, order types and delivery platforms, stock units and conversions, minimum stock levels, expiry warning days, inactivity logout, backups.	Writes: users, roles, settings, audit_logs.

4 Inventory calculation formulas

This section is the heart of the stock system. Every number on the Closing Stock screen comes from the formulas below. The developer must implement them exactly and must be able to reproduce every worked example to the cent.

4.1 The stock equation



Example: Chicken Breast. All figures are calculated by the system; the manager only types the CLOSING count.

Figure 4.1 - From opening stock to value used (example: Chicken Breast, KSh 500 per kg).

Core formulas (per product, per closing period)

```
Unit price = Total purchase price / Quantity received
Total available = Opening + Received + Adjustments IN - Adjustments OUT - Wastage
Stock used = Total available - Closing (physical count)
Value of stock used= Stock used x Weighted average unit cost
Total stock cost out = Value of stock used + Value of wastage + Value of adjustments OUT
```

Note on wastage. The brief defines Total available as Opening + Received + Adjustments in - Adjustments out. Wastage is recorded separately so management can see where stock goes, but it still leaves the shelf. To avoid counting wasted stock as "used", wastage is deducted before Stock used is calculated. The closing screen shows wastage in its own column. If the owner prefers wastage to be part of "used" instead, only the display changes; the stored figures do not.

OWNER DECISION REQUIRED

Decision: Should wasted stock be shown separately from normal use?

Recommended default: Yes. Stock used = consumption only. Wastage is a separate column and a separate report line. Reports show both, and a "Total stock cost out" line that adds them.

Owner to approve or change before development of this feature starts.

4.2 Unit price on receipt

Calculated automatically; the user never types the unit price

```
Unit price = Total purchase price / Quantity received
Example: Cooking Oil, 20 litres, KSh 6,000.00 -> 6,000.00 / 20 = KSh 300.00 per litre
```

- Store unit price with **4 decimal places** internally (e.g. a 3 kg bag at KSh 1,000 gives 333.3333 per kg) and display 2 decimals.
- The quantity must be greater than 0 and the total price must be 0 or more. A zero price is allowed (free sample) but the system shows a confirmation prompt.
- Quantity and price are stored exactly as entered. The system never changes a saved receipt; corrections are made through audited adjustments (Section 4.8).

4.3 Costing method: weighted average (recommended)

A product is often bought at different prices on different days. The system needs one cost per unit to value the stock used. The recommended method for version 1 is **weighted average cost**, because it is simple, fair, and easy to explain.

Weighted average, calculated once per product at closing

```
Opening value = Opening quantity x Opening average cost (from last closing)
Received value = sum of (each receipt's total purchase price in this period)
Adjustments IN value = Adjustment quantity x current average cost (or cost entered)
Pool quantity = Opening qty + Received qty + Adjustments IN qty
Pool value = Opening value + Received value + Adjustments IN value
Average unit cost = Pool value / Pool quantity
Closing value = Closing quantity x Average unit cost (becomes next Opening value)
```

Supplier returns reduce Received quantity and Received value at the credited amount (they are treated as negative receipts). **Adjustments OUT, wastage and stock used** are all valued at the same average unit cost.

Worked example 1 - single price (matches the brief)

Item	Value
Opening stock	10 kg
Received	5 kg at KSh 500 (KSh 2,500)
Total available	15 kg
Closing stock (counted)	4 kg
Stock used	15 - 4 = 11 kg
Average unit cost	KSh 500.00 per kg
Value of stock used	11 x 500 = **KSh 5,500.00**

Worked example 2 - two different purchase prices

Cooking Oil: 12 L were left from yesterday, valued at KSh 290 per litre (KSh 3,480). Today 20 L arrived at KSh 6,000 (KSh 300 per litre). The manager counts 21 L at closing.

Step	Calculation	Result
Pool quantity	12 + 20	32 L
Pool value	3,480.00 + 6,000.00	KSh 9,480.00
Average unit cost	9,480.00 / 32	KSh 296.25 per L
Stock used	32 - 21	11 L
Value of stock used	11 x 296.25	**KSh 3,258.75**
Closing value (next opening)	21 x 296.25	KSh 6,221.25

Worked example 3 - with wastage

Tomatoes: opening 8 kg at KSh 110 (KSh 880), received 15 kg for KSh 1,800 (KSh 120 per kg), 1 kg thrown away as spoiled, closing count 6 kg.

Step	Calculation	Result
Pool quantity / value	8 + 15 ; 880 + 1,800	23 kg ; KSh 2,680.00
Average unit cost	2,680 / 23	KSh 116.52 per kg (116.5217 stored)
Total available	8 + 15 + 0 - 0 - 1 (waste)	22 kg
Stock used	22 - 6	16 kg
Value of stock used	16 x 116.5217	**KSh 1,864.35**
Value of wastage	1 x 116.5217	KSh 116.52

Rounding rule

Keep unit costs to 4 decimal places during calculation. Round each line's money value to 2 decimal places (half up) once, at the end. Report totals are the **sum of the rounded lines**, so the printed lines always add up to the printed total.

4.4 Unit conversions

Every product has one **base unit** (e.g. kg for rice, litres for oil). All stock is stored in the base unit. Users may *enter* a quantity in another compatible unit and the system converts it.

Dimension	Units (factor to base)	Example
Weight	kg = 1; g = 0.001	500 g entered for a kg product is stored as 0.5 kg
Volume	L = 1; ml = 0.001	750 ml entered for a litre product is stored as 0.75 L
Count	pcs = 1; dozen = 12; bottles, packets, balls = 1	2 dozen entered for a pieces product is stored as 24 pcs
Incompatible	kg and L, pcs and kg, etc.	**Blocked.** Message: "Kilograms cannot be converted to litres for this product."

- Conversion factors live in a **units** table (dimension + factor), editable by the administrator. Developers must never hard-code conversions.
- A unit with weight-to-volume conversion (e.g. cooking oil by weight) is **not** supported in version 1.
- The unit price is stored per base unit. If 5 kg is entered as 5,000 g for KSh 2,500, the unit price is still KSh 500 per kg.

4.5 Negative stock

The system never silently allows negative inventory. Rules:

- **Closing count** cannot be higher than Total available. The screen shows: "Closing cannot be more than available (15 kg). Record the missing delivery or an adjustment first."
- **Adjustment OUT** or **wastage** cannot exceed the quantity on hand.
- **Supplier return** cannot exceed the quantity received on that purchase line.
- An authorised manager may override only through a documented stock adjustment with a reason, which is logged.

4.6 Expiry and low-stock rules

- **Low stock:** each product has a minimum level. When closing quantity (or live balance) is at or below it, the product appears in the low-stock alert on the dashboard and in the closing screen.
- **Expiry:** each purchase line has an optional expiry date. A line is **amber** when it expires within the warning period (default 5 days, configurable) and **red** when expired.
- The system does not track which physical batch is on the shelf. It assumes **first in, first out**: the oldest receipts are used first, so the stock still on hand belongs to the *newest* receipts. Expiry alerts apply only to receipt lines that, by this rule, still have stock on hand.

> **OWNER DECISION REQUIRED**
> **Decision:** How many days before expiry should the system warn, and is first-in-first-out the correct assumption for the kitchen?
> **Recommended default:** 5 days warning (adjustable per product category) and FIFO assumption. If some items are used newest-first, the owner must say so.
> *Owner to approve or change before development of this feature starts.*

4.7 Periods, not dates

A closing covers the **period** from the previous closing's submission until this one is submitted. Receipts, adjustments and wastage are assigned to the open period by their timestamps. This avoids midnight problems when the restaurant closes after midnight. A restaurant that counts stock twice a day (lunch and dinner) simply submits two closings; the formulas are the same.

> **OWNER DECISION REQUIRED**
> **Decision:** Is stock counted once a day or at the end of every shift?
> **Recommended default:** Once per day at close, with the setting "closings per day" left open so shift closings can be enabled without code changes.
> *Owner to approve or change before development of this feature starts.*

4.8 Corrections to stock

- Purchase records are never edited or deleted. A mistake (wrong quantity, wrong price) is fixed by an **authorised stock adjustment** (IN or OUT, with the cost if a price was wrong) and a written reason.
- Each adjustment stores: product, direction, quantity, unit cost, reason, user, optional approver, date/time. It appears in the audit log and in the closing report.
- A submitted closing is locked. See Section 5.4 for the correction path.

5 Stock-taking workflows

5.1 Closing process

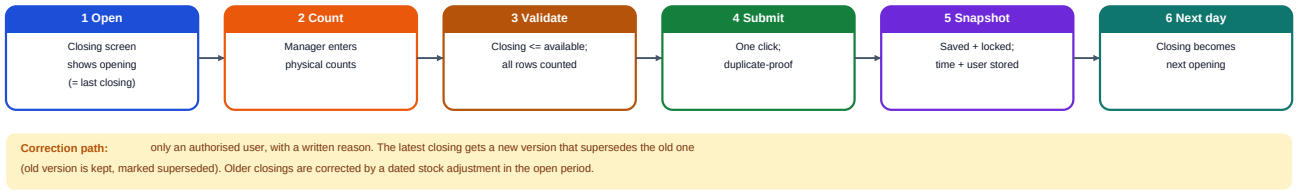


Figure 5.1 - Closing stock lifecycle.

5.2 What the closing screen shows

The screen has two separate groups, each in **alphabetical order**: **A. Purchased inventory** and **B. Makeline inventory**. Only the Closing column accepts input; everything else is calculated.

Purchased inventory column	Source / calculation
Product name, unit	products table (base unit)
Opening stock	Closing quantity of the previous submitted closing (first ever closing: entered once during setup)
Purchased stock received	Sum of receipt quantities in the open period (net of supplier returns), in base unit
Adjustments, Waste	Sum of authorised adjustments (IN minus OUT) and wastage records in the open period
Total available stock	Opening + Received + Adjustments IN - Adjustments OUT - Wastage
Closing stock (input)	Physical count typed by the manager. Must be 0 or more and not above Total available
Stock used	Total available - Closing
Unit cost	Weighted average (Section 4.3)
Value of stock used	Stock used x Unit cost, rounded to 2 decimals

Makeline column	Source / calculation
Product name, unit	makeline_products table
Opening quantity	Closing quantity of previous closing
Quantity produced	Sum of makeline production in the open period
Total available quantity	Opening + Produced + adjustments IN - adjustments OUT - wastage
Closing quantity (input)	Physical count
Quantity used	Total available - Closing

No cost or value is calculated for makeline products in version 1.

5.3 Submit rules

1. Every product row must have a closing count. A count of 0 must be typed (blank is not 0). Rows left blank block submission and are highlighted.
2. Validation (Section 4.5) runs first. Errors are shown beside the row, not as a pop-up.
3. The button is disabled after the first click and the request carries a unique **idempotency key**, so a double click or a retry after a poor connection cannot create two closings.
4. The system saves the session and its line items in **one database transaction**. If any part fails nothing is saved and the manager sees a clear message.
5. On success the closing is marked **SUBMITTED** and **locked**. The closing quantities become the next opening quantities and the closing values become the next opening values.
6. Date, time and the manager's name are stored. The closing appears in Stock History straight away.

5.4 Corrections after submission

- A submitted closing cannot be edited by Manager or Cashier users.

- An authorised user (Manager with permission, or Administrator) may correct the **latest** closing. The system creates a **new version** that links to the original (supersedes), stores a written reason, and marks the old version SUPERSEDED. Both stay visible in Stock History.
- Older closings are never re-opened (that would change every later opening stock). Instead, a dated **stock adjustment** with a reason is entered in the current period.
- The system never overwrites a previous closing record.

OWNER DECISION REQUIRED

Decision: Who may correct a closing, and how far back?

Recommended default: Manager (with the "correct closing" permission) may correct the latest closing only; the Administrator can also approve adjustments older than that. Every correction needs a reason and is logged.

Owner to approve or change before development of this feature starts.

5.5 The very first closing (setup)

Before the first closing there is no opening stock. During setup the manager performs an **opening stock count** for every product and makeline item, entering quantity and cost per unit (from the latest invoices). This is saved as an "Opening balance" closing marked as setup. From then on the normal workflow applies.

6 Makeline workflow

Makeline items are things the kitchen **makes** before selling or using them: prepared chicken, sauces, marinated meat, chopped vegetables, dough, cooked sushi rice. They are separate from purchased ingredients everywhere in the system.

6.1 Rules

- Makeline products live in their own table, with their own screen, list and closing section.
- Each production record stores: product, quantity, unit, date and time, person responsible, optional notes. Records are kept as history and not deleted.
- Makeline stock is **not** part of the financial cost-utilised calculation in version 1. No unit cost is stored for it.
- Wastage and adjustments are allowed. Both need a reason, a user and a time.
- Makeline stock is counted at every closing (Section 5.2).
- Available now = Opening + Produced - Wastage +/- Adjustments. Consumption is only known at closing (Used = Available - Closing) because version 1 has no recipes.

6.2 Building for the future (recipes)

Later, a recipe will say, for example, that 1 kg of Marinated Chicken uses 1.05 kg chicken breast, 0.05 L oil and 0.02 kg spices. To make that possible without rebuilding anything now:

- Create the **recipe_lines** table now (empty). Columns: output (menu item or makeline product), input (product or makeline product), quantity, unit.
- Keep purchased and makeline products in separate tables with the same stock-movement structure, so a recipe can point at either.
- Add a **recipe_deducted** flag to order lines (false in version 1) so that, in Phase 4, sales can deduct ingredients automatically without double-counting older orders.
- Do **not** assume every menu item has a recipe. Many will never have one.

OWNER DECISION REQUIRED

Decision: Should "Total available" for makeline items allow wastage and adjustments to be recorded at any time?

Recommended default: Yes. Recorded immediately, always with a reason, visible in the closing screen the same way as for purchased stock.

Owner to approve or change before development of this feature starts.

7 Menu management

Menu Management defines everything the restaurant sells. Items saved here appear in the POS immediately. Management can create as many menu items and categories as needed.

Field	Required	Notes
Item name	Yes	Shown in POS, kitchen ticket and receipt. Up to 60 characters.
Selling price (KSh)	Yes	Stored with 2 decimals. Interpreted as VAT-inclusive or VAT-exclusive according to the global VAT setting (Section 8.5).
Category	Yes	Starters, Main Meals, Milkshakes, Sushi, Teas, Mocktails, Desserts, Sides, Specials, and any the owner adds.
Description	No	Short text, shown on the management screen and optionally on tablet POS.
Image	No	JPG/PNG up to 2 MB, resized by the system. POS works with a plain colour tile if no image exists.
Availability status	Yes	Available / Unavailable today (e.g. sold out). Unavailable items are greyed out in POS.
Active / inactive	Yes	Inactive items are hidden from POS but stay in history and reports.
VAT setting	Yes	Standard / exempt / zero-rated, following the global VAT table. The rate itself is never typed on the item.
Item code	No	Optional short code or barcode for fast search (e.g. SUS-004). Unique when entered.

7.1 Functions

- Add new items and categories; edit any field; reorder categories.
- Edit prices. Each price change writes a row to **menu_item_price_history** (old price, new price, who, when, optional reason).
- Change an item's category; activate or deactivate; mark unavailable temporarily.
- Search by name or code; filter by category; see inactive items on request.
- Never delete an item that has been sold. The delete button is replaced by **Deactivate**.

7.2 Price and history rules

- A price change applies to **new orders only**. Every order line stores the item name, price and VAT treatment at the moment of sale, so old orders never change.
- Only Manager and Administrator can change prices. The cashier cannot change an item price in POS; a discount is a separate, authorised action.

OWNER DECISION REQUIRED

Decision: Are menu prices shown to customers VAT-inclusive (the usual practice for restaurants) or VAT-exclusive?

Recommended default: VAT-inclusive. Set once in Settings; the POS then shows the VAT portion contained in the total.

Owner to approve or change before development of this feature starts.

8 POS and order processing

The POS is used by cashiers during service, so it must be fast, large-button, tablet-friendly and forgiving. Layout (see Screen 7): order-type chips along the top, categories on the left, a searchable grid of menu items in the middle, and the cart on the right.

8.1 Order types

Every order has exactly one order type, chosen first. Version 1 ships with eight: **Dine In, Pick Up, Glovo, Bolt Food, Uber Eats, SpeedyDrop, NAI Deliver, Yum Deliveries**. The administrator can add new delivery platforms or deactivate existing ones in Settings; deactivated types disappear from POS but remain in history.

Order type	Extra details to capture	Typical payment
Dine In	Table number (required). Customer name optional.	Cash, M-Pesa, card, split
Pick Up	Customer name (required), phone optional.	Cash, M-Pesa, card
Delivery platforms (6 + any added)	Platform order reference (required). Rider name optional.	Usually "Platform payout" - paid by the platform to the restaurant later

8.2 Data stored on every order

Group	Fields
Identity	Unique order number (e.g. ORD-20261106-0042: date + daily sequence) and a system unique identifier (UUID). Date and time, created automatically.
Type and people	Order type; cashier / employee; optional table number, customer name, customer phone, delivery reference.
Items	For each line: menu item (and its name at that time), quantity, individual selling price, line total, line discount, VAT amount, special instructions.
Totals	Subtotal, discount, VAT, grand total, VAT mode used (inclusive / exclusive) and VAT rate used (stored on the order).
Payment	Payment method(s), payment status (unpaid / partially paid / paid / refunded), M-Pesa or card references.
Status	Order status (new / preparing / ready / served or collected / cancelled), void or refund flag with reason, timestamps for each status change.

8.3 POS order workflow

1. Employee selects the **order type** (and enters table number or delivery reference).
2. Employee searches for an item or taps a category.
3. Employee taps items to add them to the cart; tapping again adds one more.
4. Employee adjusts quantities with + and - buttons, and adds a note (e.g. "no onions").
5. System calculates subtotal, discount, VAT and total **automatically** after every change.
6. Employee presses **Complete order** (or **Send to kitchen** first for dine-in orders that will be paid later).
7. System generates a unique order number.
8. System saves the order, its lines and its payment(s) in one database transaction.
9. System creates a kitchen ticket (if the order type or item category requires one).
10. System prints or shows the customer receipt.
11. Order is stored permanently and appears in sales history.

8.4 Protection against duplicate or lost orders

- The cart is given a unique **idempotency key** when it starts. Submitting the same key twice returns the original order instead of creating another.
- The Complete button is disabled while the request is in progress.
- If the connection fails during submit, the screen says "Not saved - check connection" and keeps the cart. Nothing is shown as sold until the server confirms.

- Order numbers are produced by the server using a locked counter so two cashiers can never receive the same number.

OWNER DECISION REQUIRED

Decision: Should the POS keep working when the internet is down (offline mode)?

Recommended default: Version 1 requires a working local connection to the system. A simple offline queue is a Phase 4 option. Meanwhile the owner should keep a backup router/mobile hotspot for the POS tablet.

Owner to approve or change before development of this feature starts.

8.5 VAT calculation (configurable)

Administrators control: VAT enabled or disabled, the VAT rate, whether menu prices include or exclude VAT, and which items are VAT-eligible. Rates are stored in the **vat_settings** table with an effective date; the system never hard-codes a rate. Each order stores the rate and mode it used, so a later rate change does not alter old orders.

VAT formulas (sample rate 16% - configurable)

```
VAT-INCLUSIVE prices (recommended): VAT = Line total x rate / (100 + rate) Net = Line total - VAT
VAT-EXCLUSIVE prices: VAT = Line net x rate / 100 Total = Net + VAT
Discounts are applied to the line BEFORE VAT is worked out.
Round VAT per line to 2 decimals (half up). Order VAT = sum of line VAT.
Example (inclusive, sample rate 16%): KSh 4,750.00 x 16 / 116 = KSh 655.17 VAT; Net = KSh 4,094.83
```

Items marked exempt or zero-rated carry VAT of 0 on their line. The receipt shows net, VAT and total; reports show "VAT collected" for any date range.

8.6 Discounts, voids and refunds

- **Discounts** need the "apply discount" permission (Manager/Admin by default). The cashier chooses amount or percentage and a reason code; the discount appears on the receipt and in the audit log.
- **Completed orders are never deleted.** An authorised user may **void** an order or **refund** part or all of it. A reason is mandatory, the original order stays visible, and a refund creates negative payment records.
- A void or refund before the order is served may also cancel the kitchen ticket.

8.7 Payments

- Payment methods: Cash, M-Pesa, Card, Bank transfer, plus configurable others (e.g. Platform payout). Each method can require a reference.
- **Payment status**: Unpaid, Partially paid, Paid, Refunded.
- **Split payments:** an order can have several payment rows (e.g. KSh 2,000 cash + KSh 2,750 M-Pesa). The order is Paid when the payments equal the grand total.
- Cash: the screen asks for "amount tendered" and shows change due. Change is not stored as revenue.
- M-Pesa in version 1: the cashier types the transaction code. Automatic M-Pesa confirmation (Daraja API) is a Phase 4 option.

8.8 Sales history

Every completed order is stored permanently. Management can search by **date, order number, menu item, order type, employee, payment method and payment status**, open any past order to see exactly what was sold, reprint a receipt (marked "Reprint"), and view its audit trail.

9 Kitchen ticket workflow

When an order is confirmed, the system creates a **kitchen ticket**. It shows the order number, order type, date and time, table number (if dine-in), item names and quantities, special instructions and notes.

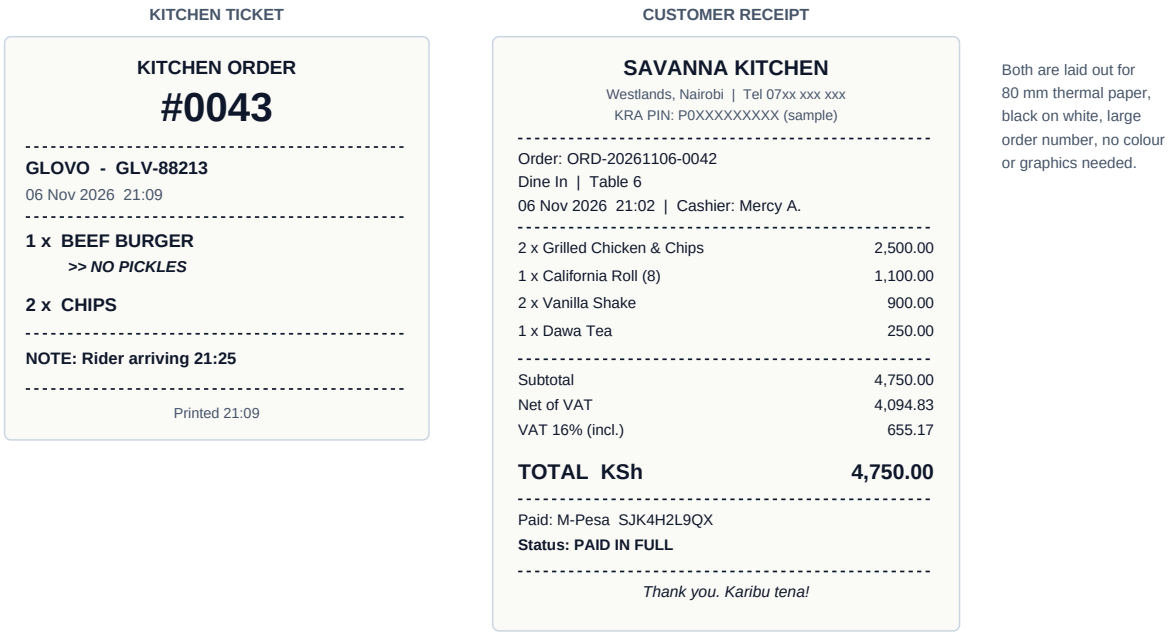


Figure 9.1 - Thermal ticket and customer receipt (80 mm). Sample restaurant name and PIN.

9.1 Status flow

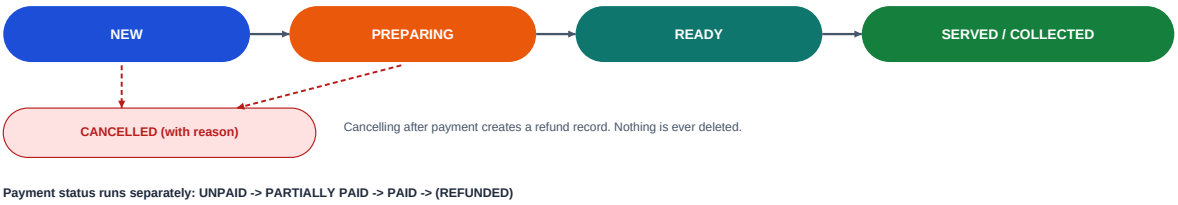


Figure 9.2 - Kitchen status flow and cancellation.

Status	Who changes it	What happens
New	System (on order confirmation)	Ticket appears in the kitchen's "New" column; prints if auto-print is on.
Preparing	Kitchen staff	Tap "Start preparing". POS shows "Preparing" next to the order.
Ready	Kitchen staff	Tap "Mark ready". POS shows a highlighted "Ready" so the waiter or rider is called.
Served / Collected	Cashier or kitchen	Dine-in: "Served". Pick-up and delivery: "Collected". Ticket leaves the board.
Cancelled	Manager (reason required)	Ticket struck through on the board; a reprint says CANCELLED.

- Every status change stores who and when, so average preparation time can be reported later.
- The kitchen screen refreshes automatically (every few seconds or by live push) and plays a short sound when a new ticket arrives.
- A ticket can be printed again; the reprint counter is shown on the ticket ("Reprint #1").
- Station routing (e.g. sushi bar vs hot kitchen) is a later option. Version 1 prints one ticket per order.

OWNER DECISION REQUIRED

Decision: Which orders need kitchen tickets, and how many printers are there?

Recommended default: All orders with food items print one ticket; drinks-only orders do not. One thermal printer in the kitchen. Receipts print at the cashier's printer. The owner must confirm the printer models (80 mm ESC/POS recommended).

Owner to approve or change before development of this feature starts.

10 Reporting requirements

10.1 Dashboard

The dashboard (Screen 1) is the owner's five-second view of the restaurant. It shows today's sales, number of orders, average order value, sales by order type and payment method, best-selling items, stock value received today, stock used value (from the last closing), wastage, low-stock items, and products approaching expiry.

10.2 Required reports

Report	Filters	Key columns
Daily / weekly / monthly sales	Date range	Orders, gross sales, discounts, VAT, net sales, average order
Sales by menu item	Date, category	Item, quantity sold, sales value, share of total
Sales by category	Date	Category, quantity, sales value
Sales by delivery platform	Date, platform	Orders, sales, average order, share of sales
VAT collected	Date	Taxable sales, VAT, exempt sales (supports the VAT return)
Stock received	Date, supplier, product	Date, product, quantity, total price, unit price, invoice
Stock used	Date range, product	Opening, received, used, closing, value used
Wastage	Date, reason, employee	Product, quantity, value, reason, who
Closing stock history	Date, compare two dates	Full closing snapshot per day
Employee sales activity	Date, employee	Orders, sales, discounts, voids and refunds

All reports can be exported to **PDF** (print-ready, with restaurant name, date range and who ran it) and **Excel** (.xlsx, one table per sheet, numbers as numbers not text).

10.3 Keep the money terms separate

Sales revenue, stock used and gross profit are three different things

Sales revenue = money from customers (gross, then net of VAT). **Stock utilisation value** = what the stock used in the period cost the restaurant. **Gross profit** = revenue (net of VAT) minus the cost of what was sold.

In version 1 the system has no recipes, so it cannot know the cost of each sale. "Stock used value" is therefore shown as **stock utilisation**, never as net profit and never as gross profit. Reports must label it exactly "Stock used value" and show a note: "Not profit. Excludes wages, rent, utilities and other expenses." A true gross profit and food-cost percentage arrive in Phase 4 with recipes.

11 User roles and permissions

There are four simple roles. A user has exactly one role. The Administrator can create users and reset passwords. Permissions are stored as codes in a **permissions** table linked to roles, so the owner can later allow or block individual actions without code changes.

Role	Purpose	Can do
Administrator / Owner	Runs the system	Everything: settings, users, VAT, reports, approvals, audit log, backups.
Manager	Runs daily operations	Inventory received, makeline, closing stock, menu management, reports, authorised adjustments, discounts, voids/refunds (if permitted).
Cashier	Takes orders	POS, orders, payments, receipts, view own orders, mark orders served/collected.
Kitchen Staff	Prepares food	View kitchen tickets and update preparation status. Record makeline production (if permitted).

Permission	Admin	Manager	Cashier	Kitchen
View dashboard (full)	Yes	Yes	-	-
Receive stock / supplier returns	Yes	Yes	-	-
Edit suppliers	Yes	Yes	-	-
Record makeline production	Yes	Yes	-	Option
Record wastage	Yes	Yes	-	Option
Stock adjustment (needs reason)	Yes	Yes	-	-
Enter and submit closing stock	Yes	Yes	-	-
Correct submitted closing	Yes	Option	-	-
View stock history	Yes	Yes	-	-
Add / edit menu items and prices	Yes	Yes	-	-
Use POS, take payments, print receipts	Yes	Yes	Yes	-
Apply discount	Yes	Yes	Option	-
Void / refund an order	Yes	Yes	-	-
View all sales history	Yes	Yes	Own	-
View kitchen tickets, update status	Yes	Yes	View	Yes
View / export reports	Yes	Yes	-	-
Manage users, roles, VAT, settings	Yes	-	-	-
View audit log	Yes	Own area	-	-

"Option" = the administrator can switch this permission on or off for the role. "Own" = only the user's own records.

11.1 Security requirements

- **Secure login** with username/email and password. Passwords stored as salted hashes (bcrypt or Argon2), never in plain text. Minimum 8 characters; lock for 15 minutes after 5 failed attempts.
- **Automatic logout** after inactivity (default 15 minutes, configurable). The POS may use a short PIN to resume a cashier session without full re-login (Phase 2 option).
- **Permission checks on the server** for every action, not only hidden buttons.
- **Audit trail** of important actions: login/logout, receipts, adjustments, wastage, closings and corrections, price changes, discounts, voids/refunds, user and settings changes. Each entry records user, time, action, record, before/after values and reason. Audit logs cannot be edited or deleted from the application.
- All traffic over HTTPS. Business data inaccessible without login. Users can be disabled (never deleted) when staff leave, so their history stays intact.

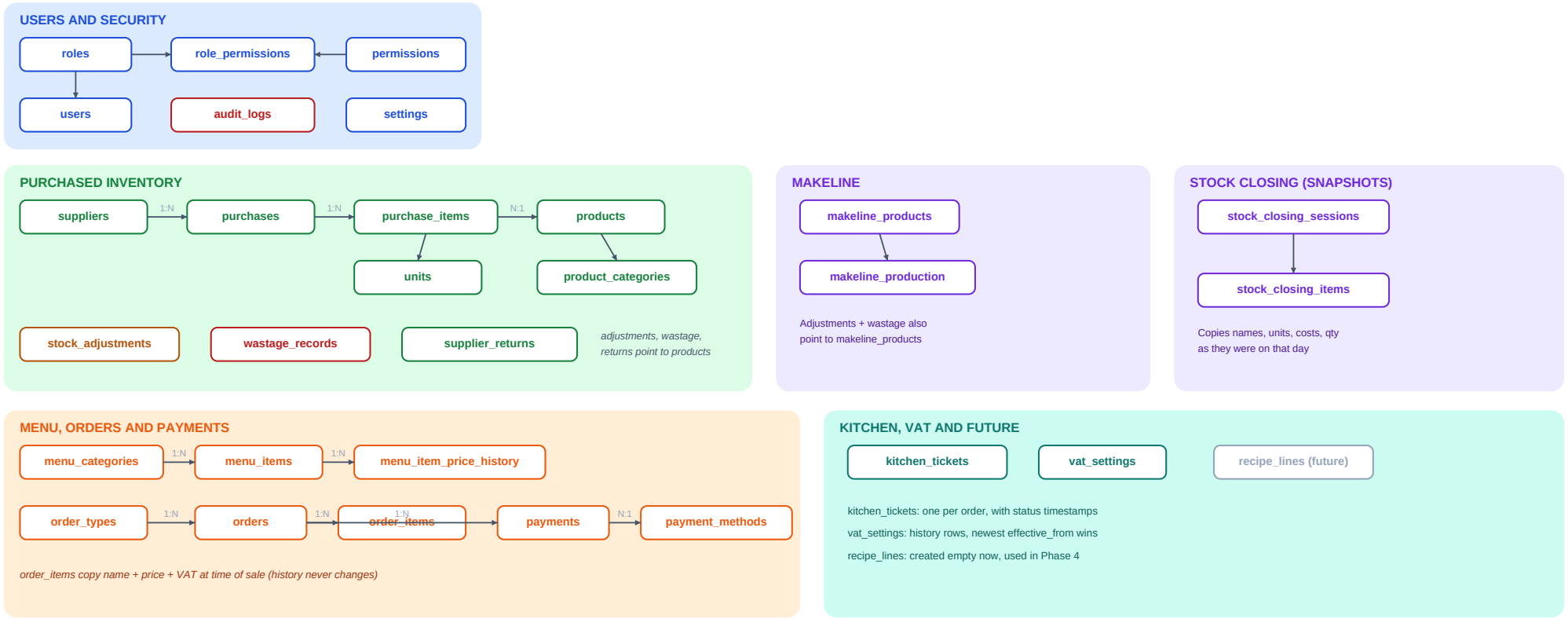
12 Database structure and technical requirements

The structure below is intentionally simple: about 30 tables, standard relational design, any mainstream database (PostgreSQL or MySQL/MariaDB recommended). Field lists show the important columns; the developer will add normal housekeeping columns. **PK** = primary key, **FK** = foreign key.

12.1 How the data connects

- A **purchase** (delivery) has many **purchase_items**; each item points to a **product**, a **unit** and the purchase's **supplier**.
- A **stock_closing_session** has many **stock_closing_items**. Each item *copies* the product name, unit, costs and quantities, so a closing never changes even if the product is renamed or repriced later.
- An **order** has many **order_items** and many **payments**, and one **kitchen_ticket**. Order items copy the menu item name, price and VAT at the time of sale.
- Adjustments, wastage and production records point to a **product** or a **makeline_product** (exactly one of the two).
- Every important table has **created_at** and **created_by** (users.id). The **audit_logs** table records who changed what.

Figure 12.1 - Table relationship map (arrows point from the child table to its parent).



Every transactional table carries created_at, created_by (users.id). audit_logs records who changed what, before / after, and why.

Arrows read from the "many" side to the "one" side (foreign key points to the parent record).

12.2 Tables and key fields

Users and security

Table	Key fields	Notes
users	id PK, full_name, username/email (unique), phone, password_hash, role_id FK, is_active, last_login_at, created_at	Disable, never delete.
roles	id PK, name, description	Administrator, Manager, Cashier, Kitchen Staff.
permissions	id PK, code (e.g. closing.submit), description	One row per allowed action.
role_permissions	role_id FK, permission_id FK (composite PK)	Which role has which permission.
system_settings	key PK, value, updated_by, updated_at	Business name, KRA PIN, inactivity timeout, expiry warning days, closings per day.
audit_logs	id PK, user_id FK, action, entity_type, entity_id, old_values (JSON), new_values (JSON), reason, ip_address, created_at	Insert-only. Never updated or deleted.

Purchased inventory

Table	Key fields	Notes
units	id PK, code (kg, g, L, ml, pcs...), name, dimension (weight/volume/count), factor_to_base	Drives all conversions (Section 4.4).
product_categories	id PK, name, sort_order, is_active	Oils & Fats, Dairy, Meat, Produce...
products	id PK, name (unique), category_id FK, base_unit_id FK, min_stock_level, on_hand_qty, avg_unit_cost, is_active, created_at	on_hand_qty / avg_unit_cost are cached balances; the ledger is the source of truth.
suppliers	id PK, name, contact_person, phone, email, address, kra_pin, notes, is_active	Never deleted once used.
purchases	id PK (UUID), purchase_no (unique), supplier_id FK, invoice_no, date_received, total_amount, notes, created_by, created_at	One delivery / invoice header.
purchase_items	id PK, purchase_id FK, product_id FK, quantity, unit_id FK, quantity_base, total_price, unit_price (= total_price / quantity_base), expiry_date, notes	Unit price calculated on save. Immutable.
supplier_returns	id PK, purchase_item_id FK, quantity_base, credit_amount, reason (damaged, wrong item, expired), returned_on, created_by	Reduces received quantity and value.
stock_adjustments	id PK, product_id FK or makeline_product_id FK, direction (IN/OUT), quantity_base, unit_cost, reason_code, reason_text, status, created_by, approved_by, created_at	Reason required.
wastage_records	id PK, product_id FK or makeline_product_id FK, quantity_base, unit_cost_at_time, reason_code (expired, spoiled, damaged, kitchen mistake), notes, wasted_on, employee_id FK, recorded_by, created_at	Separate from normal consumption.

Makeline

Table	Key fields	Notes
makeline_products	id PK, name (unique), unit_id FK, min_level, on_hand_qty, is_active	Separate table by design.
makeline_production	id PK, makeline_product_id FK, quantity, unit_id FK, produced_at, produced_by FK, notes, created_at	History of production.

Stock closing

Table	Key fields	Notes
stock_closing_sessions	id PK, closing_no (unique), period_start, period_end, status (DRAFT / SUBMITTED / SUPERSEDED), submitted_by, submitted_at, total_value_used, total_value_wasted, version, supersedes_id FK (nullable), correction_reason, idempotency_key (unique), created_at	Locked after submit.
stock_closing_items	id PK, session_id FK, item_type (PRODUCT/MAKELINE), product_id or makeline_product_id, name_snapshot, unit_snapshot, opening_qty, received_qty, adj_in_qty, adj_out_qty, wastage_qty, available_qty, closing_qty, used_qty, opening_value, received_value, avg_unit_cost, used_value, wastage_value, closing_value	Makeline rows leave cost fields empty (NULL).

Menu

Table	Key fields	Notes
menu_categories	id PK, name, sort_order, is_active	
menu_items	id PK, item_code (unique, nullable), name, category_id FK, price, description, image_path, is_available, is_active, vat_class_id/vat_applicable, created_at	Never deleted once sold.
menu_item_price_history	id PK, menu_item_id FK, old_price, new_price, changed_by FK, changed_at, reason	Insert-only.
recipe_lines (future)	id PK, output_menu_item_id or output_makeline_id, input_product_id or input_makeline_id, quantity, unit_id	Created now, empty, used in Phase 4.

Sales

Table	Key fields	Notes
order_types	id PK, name, is_delivery, requires_reference, is_active, sort_order	Eight to start; admin can add.
orders	id PK (UUID), order_no (unique), order_type_id FK, status, payment_status, table_no, customer_name, customer_phone, delivery_ref, subtotal, discount_total, vat_total, grand_total, vat_mode, vat_rate, cashier_id FK, idempotency_key (unique), created_at, completed_at, void_reason, voided_by	Never deleted. Voids are flags.
order_items	id PK, order_id FK, menu_item_id FK, name_snapshot, quantity, unit_price, discount, vat_rate, vat_amount, line_total, notes, recipe_deducted (default false)	Copies price and VAT at sale time.
payment_methods	id PK, name, requires_reference, is_active	Cash, M-Pesa, Card, Bank transfer, Platform payout.
payments	id PK, order_id FK, method_id FK, amount (negative for refunds), reference, status, received_by FK, created_at	Several per order allowed (split payments).
kitchen_tickets	id PK, order_id FK (unique), ticket_no, status, print_count, created_at, preparing_at, ready_at, served_at, updated_by	One per order.
vat_settings	id PK, vat_enabled, rate_percent, prices_include_vat, effective_from, changed_by FK, created_at	History kept; newest effective row applies.

12.3 Technical requirements

Area	Requirement
Unique identifiers	Orders and purchases use UUIDs plus readable numbers (ORD-YYYYMMDD-NNNN, PUR-YYYYMMDD-NNN). Numbers are issued by the server with a locked counter.
Timestamps	All important records store created_at (UTC in database, shown in East Africa Time, UTC+3). Status changes store their own timestamps.
Money and quantities	DECIMAL types only, never floating point. Money DECIMAL(14,2); quantities DECIMAL(14,4); unit cost DECIMAL(14,4).
Duplicate protection	Idempotency key on order, purchase, closing and payment submissions with a unique database constraint. Buttons disabled while submitting.
Reliable saves	Each business transaction (order + lines + payments + ticket; purchase + items + balance; closing + items + balance) runs inside one database transaction. All or nothing.

Area	Requirement
Immutability	Application database user has no DELETE right on purchases, closings, orders, payments, audit logs. Corrections create new rows.
Stock balance integrity	Cached balances (products.on_hand_qty) are updated in the same transaction as the movement. A nightly job re-computes balances from the records and flags any difference.
Backups	Automatic full backup every night and incremental every hour during trading; copies stored in two places (server + off-site/cloud). Keep 30 daily and 12 monthly backups. Backups encrypted.
Recovery	Written recovery procedure; target: restore within 4 hours, lose no more than 1 hour of data. A **restore test** must be done before go-live and every quarter.
Access protection	HTTPS, hashed passwords, role checks on the server, login rate limiting, session expiry, no database exposed to the internet, security updates applied.
Devices	Desktop and tablet first; responsive to phone widths. Works in current Chrome, Edge, Safari. Thermal printing through browser print (80 mm) or ESC/POS.
Performance	Product, menu and order searches respond in under 1 second with 5 years of data. Add indexes on dates, order_no, product_id, supplier_id.

OWNER DECISION REQUIRED

Decision: Where will the system be hosted (cloud server, in-restaurant PC, or both), and who is responsible for backups?

Recommended default: Cloud hosting in a reputable provider with a Nairobi or nearby region, daily encrypted backups, plus an owner-controlled weekly export. The developer sets this up and documents it; the owner keeps the administrator credentials.

Owner to approve or change before development of this feature starts.

13 Important business rules

These 12 rules are non-negotiable. Each has a short explanation and a test the developer must pass (Section 17).

#	Rule	What it means in the system
1	**Purchased inventory and makeline inventory remain separate.**	Separate tables, screens, closing sections and reports. Totals are never combined.
2	**Stock received increases purchased inventory.**	A saved receipt raises the product's on-hand quantity and enters the weighted-average pool.
3	**Every closing updates the next opening stock.**	Next period's opening quantity and value = this closing's count and value.
4	**Historical closing records are never overwritten.**	Corrections create a new version; the old one is marked superseded and kept.
5	**Stock used = available stock less closing stock, with adjustments properly accounted for.**	Used = (Opening + Received + Adj IN - Adj OUT - Wastage) - Closing.
6	**Stock utilisation value uses the selected costing method.**	Weighted average cost (recommended). Method is stored on each closing.
7	**Makeline stock has no financial utilisation calculation in version 1.**	Cost fields are empty for makeline rows; no value appears for them anywhere.
8	**Completed POS orders are stored permanently.**	Voids and refunds are new records with reasons. Nothing is deleted.
9	**VAT is configurable.**	Rate, inclusive/exclusive and item eligibility come from settings; each order stores what was used.
10	**All important changes are traceable to a user.**	created_by on records; audit_logs for changes, with before/after and reason.
11	**The system must not silently allow negative inventory.**	See Section 4.5. Only a documented, logged adjustment can override.
12	**Stock and sales calculations are consistent and auditable.**	A report total always equals the sum of its lines; a closing can be recalculated from its source records and must match.

Future-proofing rule

Where a menu item is linked to ingredients, design for future recipe-based automatic stock deduction. Do not assume every menu item has a recipe in the initial version. Selling an item with no recipe must always work.

14 UI mockups

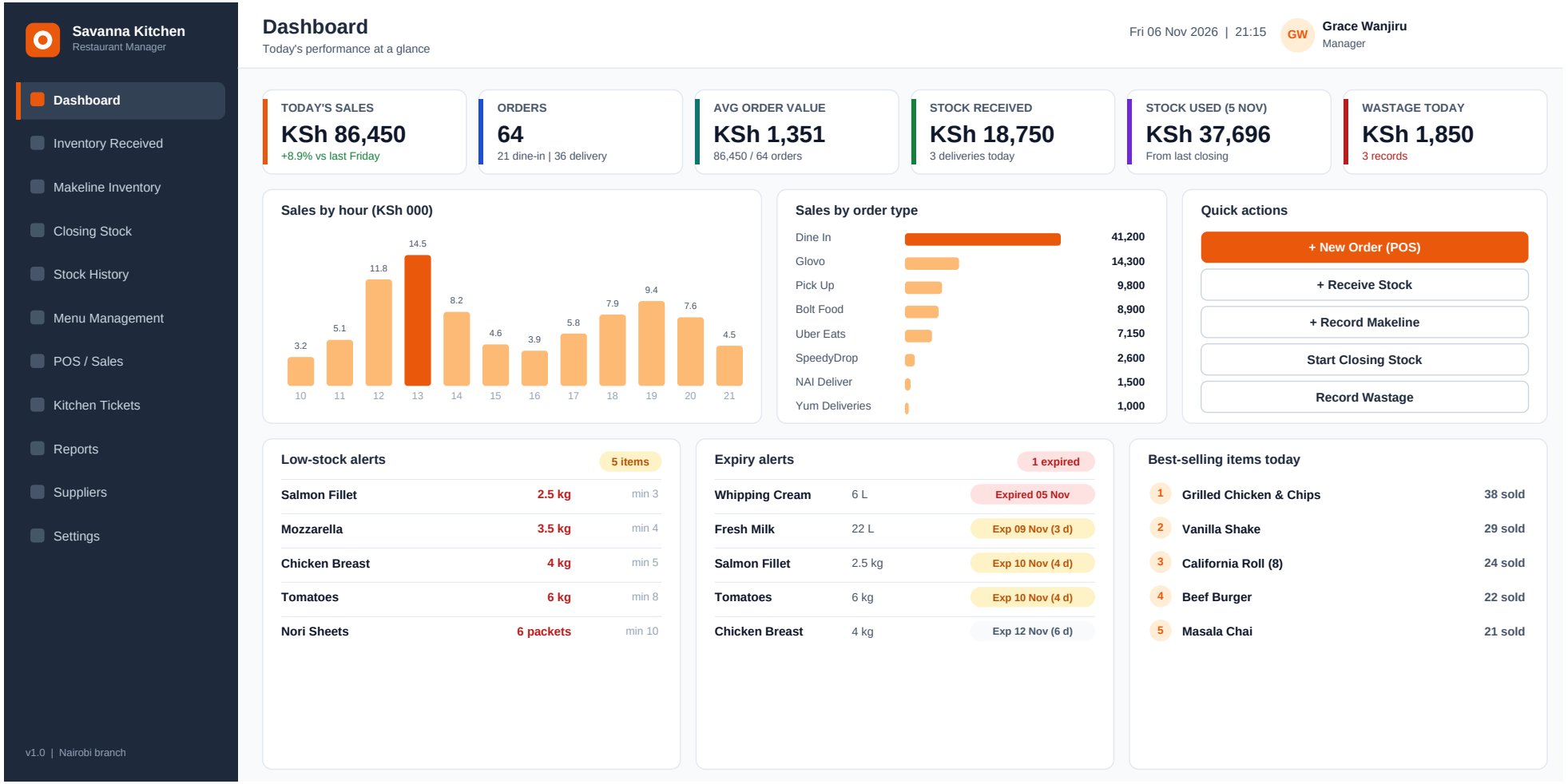
The following pages show the actual screens as they should look, with realistic sample data from Section 15. The design is a clean, modern restaurant dashboard: a dark sidebar with the 11 sections, a white top bar with the user and date, and a light content area. Desktop and tablet are the primary targets; a responsive phone layout follows the ten screens. The developer may refine colours and fonts, but the **layout, labels, order of columns and behaviours** should be kept.

14.1 Design guidelines

Item	Guideline
Colour	Dark navy sidebar; orange (#EA580C) for the single main action on a screen; green = success/paid/OK; amber = warning/expiring; red = error/expired/wastage; purple = makeline and closing; blue = information.
Typography	One clean sans-serif (e.g. Inter or system font). 14-16 px body on desktop, 16 px minimum on POS and phone. Numbers right-aligned with thousands separators.
Buttons	One primary button per screen. Destructive actions are outlined red and ask for a reason.
Forms	Required fields marked *. Calculated fields are grey/green and read-only. Errors appear next to the field in plain language.
Touch	Targets at least 44 x 44 px on tablet and phone. POS item tiles at least 120 x 110 px.
Language	English labels. Prepare text for Swahili translation later (labels stored in a language file, not typed into code).

Screen index: 1 Dashboard | 2 Inventory Received | 3 Makeline Inventory | 4 Closing Stock | 5 Stock History | 6 Menu Management | 7 POS | 8 Kitchen Tickets | 9 Reports | 10 Settings | Responsive mobile layouts.

14.2 Screen 1 - Dashboard

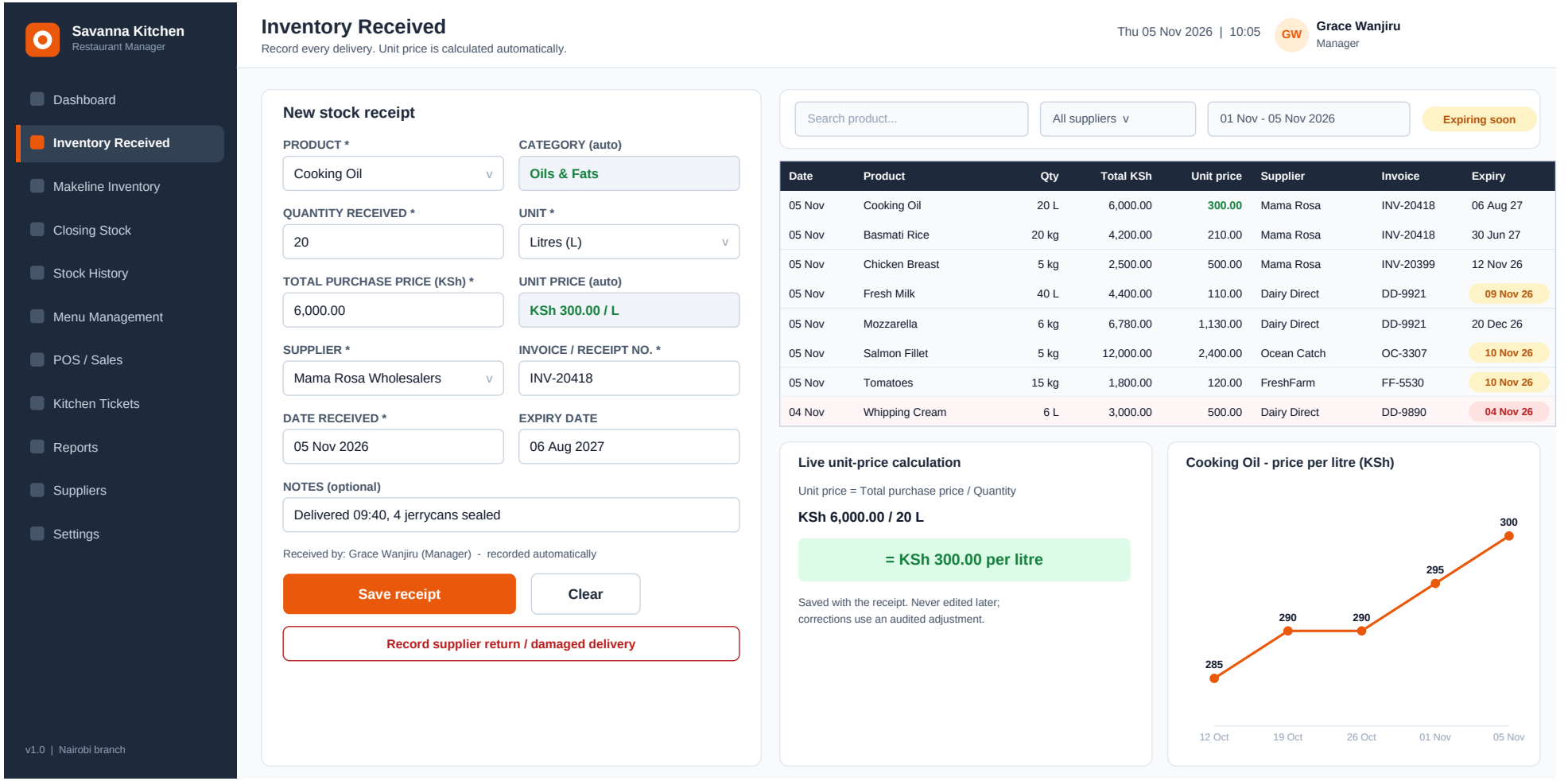


Purpose. Gives the owner or manager a five-second picture of the day. Opens by default after login for Admin and Manager.

What is on the screen. KPI cards for sales, orders, average order, stock received, stock used value and wastage. Sales by hour and by order type. Low-stock and expiry lists are clickable and open the product. Quick-action buttons jump to the five most common tasks. Stock used value is labelled with the closing date it comes from and is never called profit.

Developer notes. All figures come from saved records, not caches that can drift. Refresh every 60 seconds. Cashiers do not see this screen; they land on the POS.

14.3 Screen 2 - Inventory Received

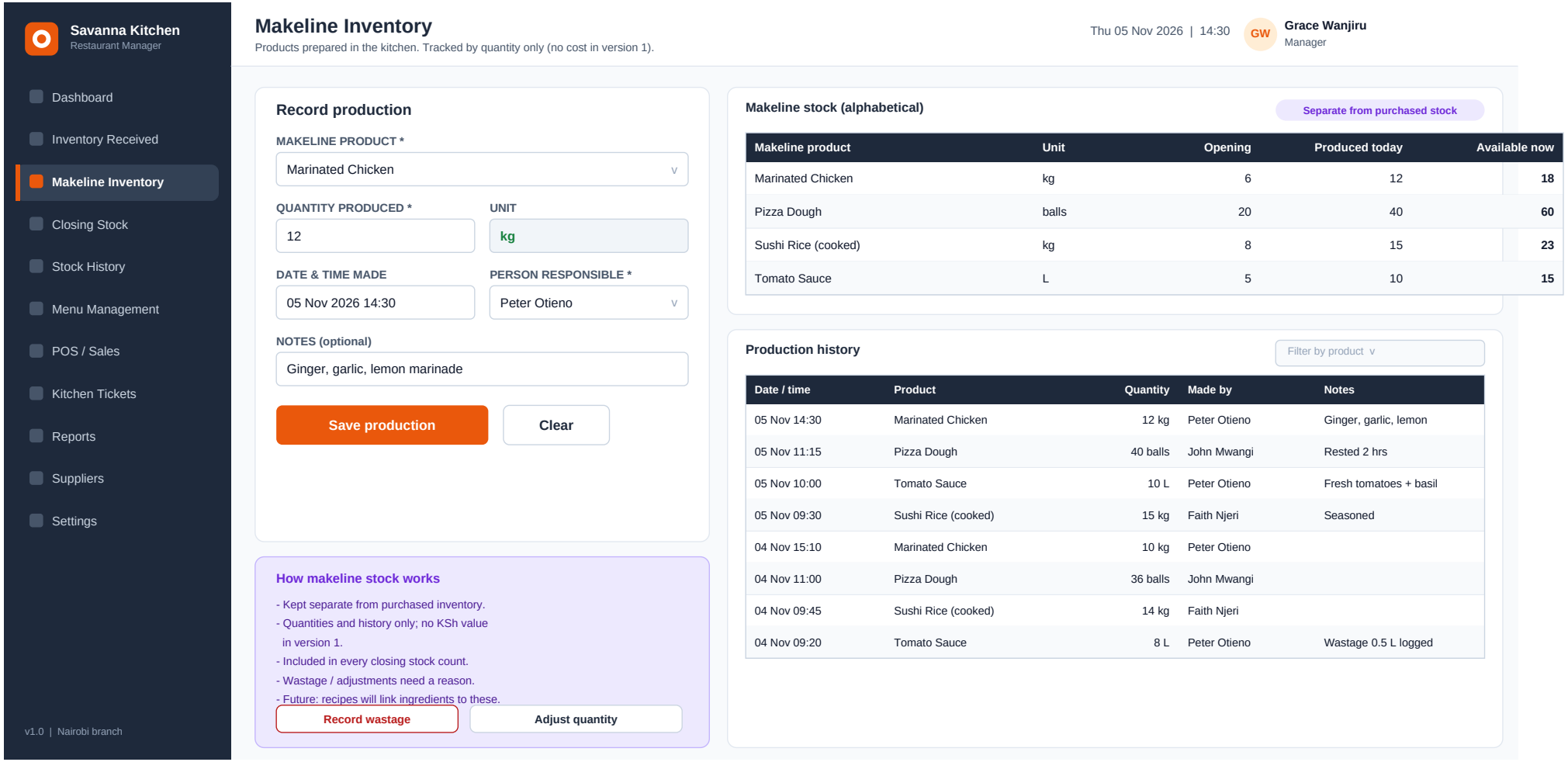


Purpose. Record each supplier delivery quickly and keep a searchable purchase history.

What is on the screen. Left: entry form. Unit price is calculated live and shown read-only. Category fills in automatically from the product. Right: purchase history with search, supplier and date filters; expiry cells turn amber when close and red when expired. Below: the live calculation and a price trend for the selected product. A separate button records supplier returns and damaged deliveries.

Developer notes. Saved receipts cannot be edited or deleted; corrections use adjustments. Received-by is the logged-in user and cannot be typed. Several products from one invoice can be added to one purchase header ("Add another item").

14.4 Screen 3 - Makeline Inventory

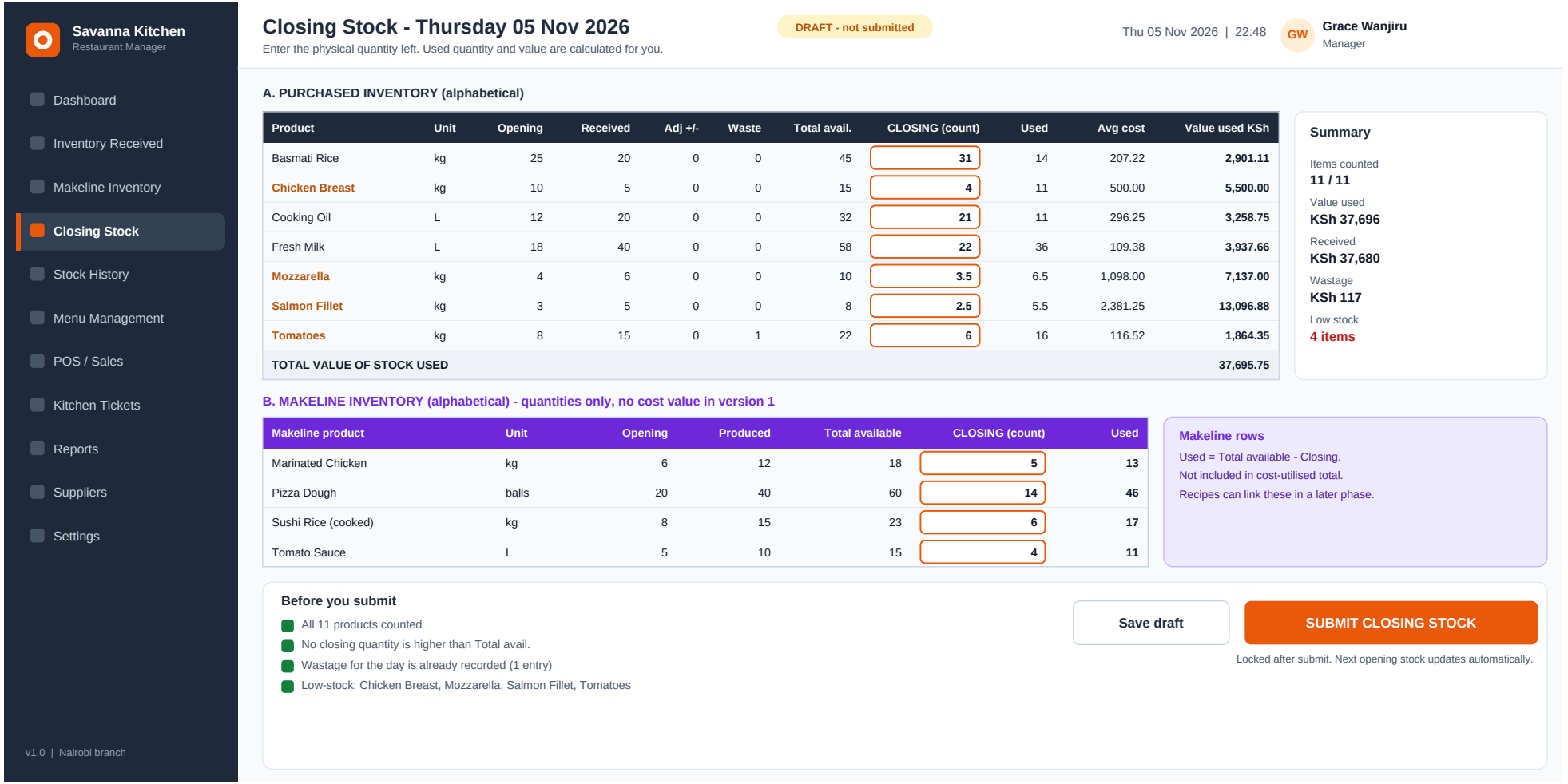


Purpose. Record what the kitchen prepares and see makeline stock separately from purchased stock.

What is on the screen. Form on the left (product, quantity, unit, date/time, person, notes). Top right: makeline stock list with opening, produced today and available now. Bottom right: production history, filterable by product. Buttons for wastage and adjustment (both require a reason).

Developer notes. No money values anywhere on this screen in version 1. Date/time defaults to now but can be set earlier for late entries (flagged as back-dated in the audit log).

14.5 Screen 4 - Closing Stock

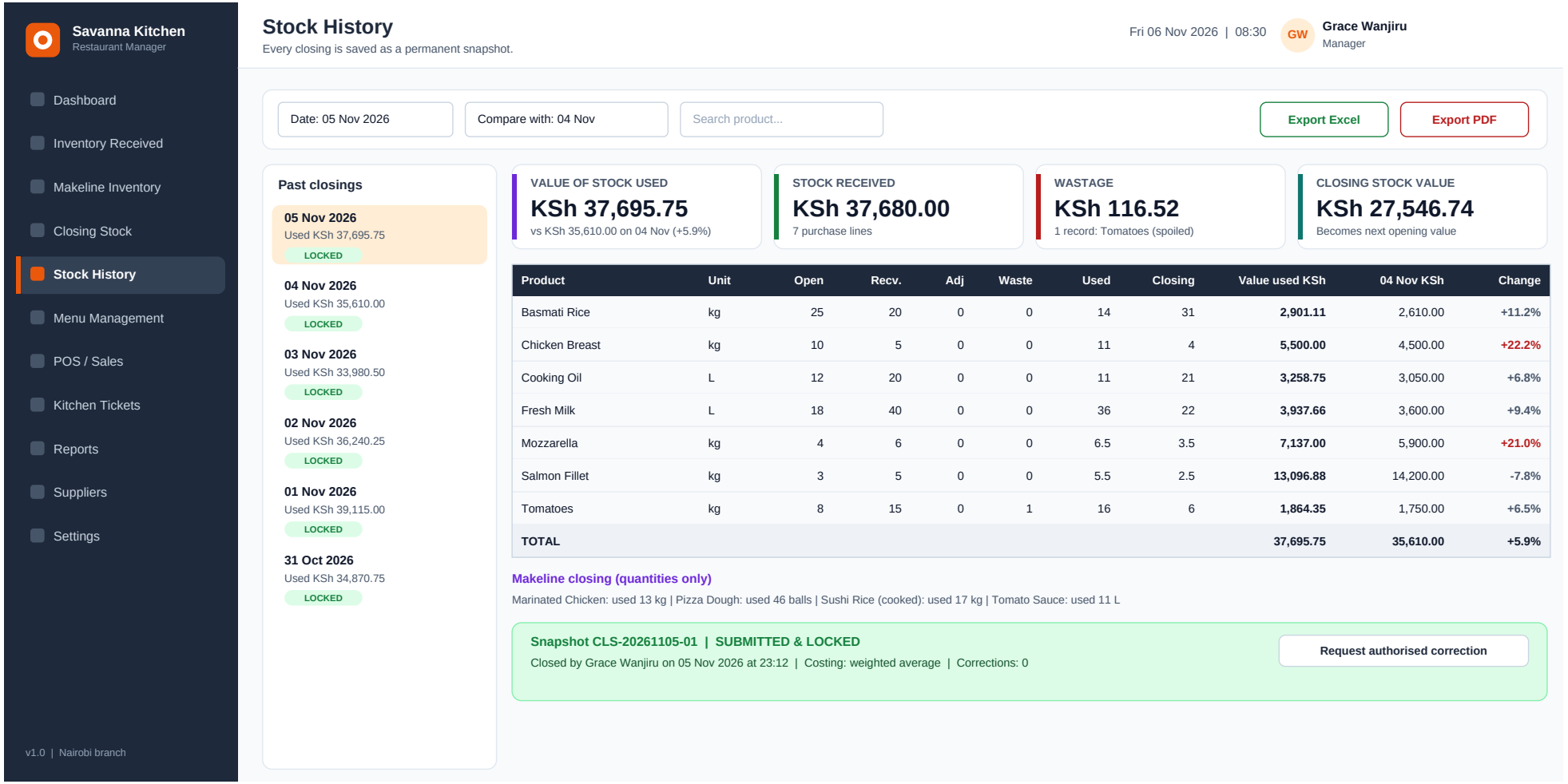


Purpose. The manager enters the physical count at the end of the day. This is the most important screen in the system.

What is on the screen. Section A lists purchased products alphabetically; Section B lists makeline products alphabetically. Only the orange CLOSING boxes accept typing. Used quantity, average cost and value used appear instantly. Product names turn amber if the closing count is at or below the minimum level. The summary panel shows totals; the footer lists what must be true before Submit.

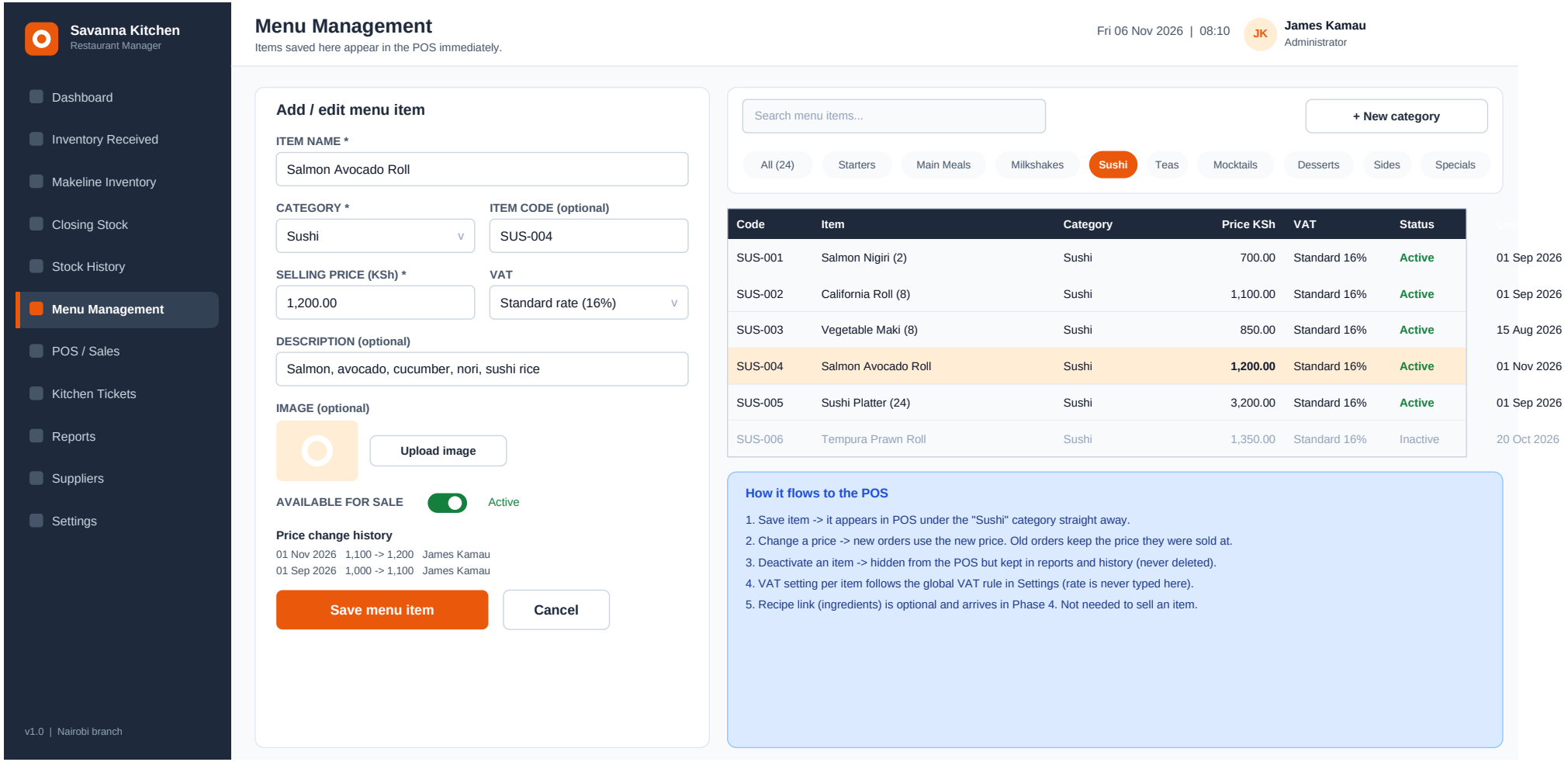
Developer notes. Submit is blocked if any row is blank or any closing exceeds Total available. After submit the screen becomes read-only and the next opening values are set. On a phone the table becomes one card per product (see mobile layouts).

14.6 Screen 5 - Stock History



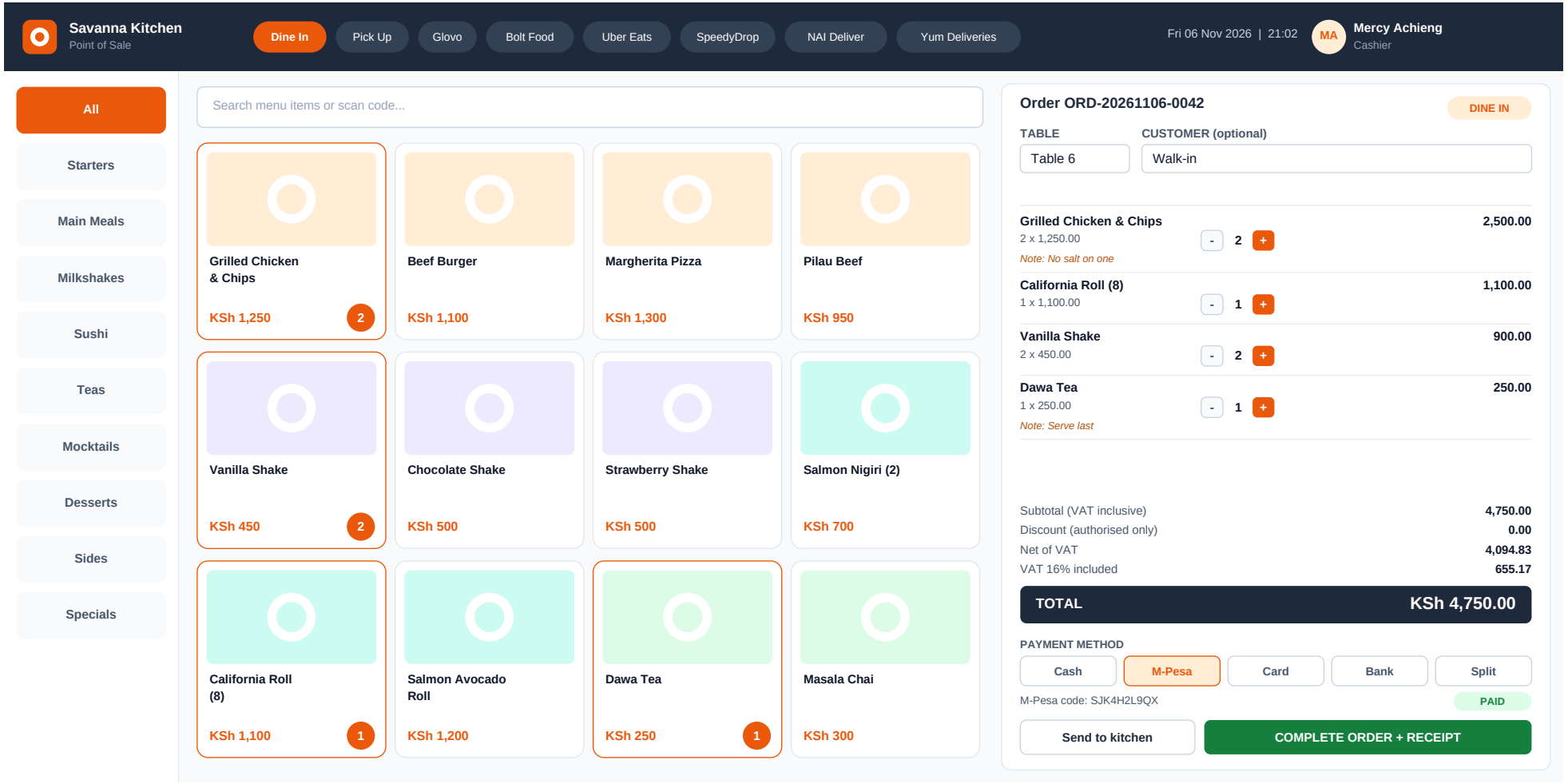
Purpose. Review any past closing as it was saved, and compare days.	**What is on the screen.** Choose a date and a comparison date. The list on the left shows past closings and their locked status. Tiles summarise value used, received, wastage and closing value. The table shows the day's movement per product and the change against the comparison day. Export buttons produce Excel and PDF. A "Request authorised correction" button appears only for permitted users and only on the latest closing.	**Developer notes.** Reads from the saved closing snapshot, never recalculates from current data. Changes above 15% are highlighted so unusual usage is noticed.

14.7 Screen 6 - Menu Management



Purpose. Create and maintain everything the restaurant sells.	**What is on the screen.** Left: add/edit form with name, category, code, price, VAT, description, image, availability and price history. Right: searchable list with category filter tabs. Selecting an item opens it in the form. Inactive items are greyed out.	**Developer notes.** Price changes write to price history. Deleting is replaced by Deactivate. Saving makes the item available in the POS at once, with no other step.

14.8 Screen 7 - POS

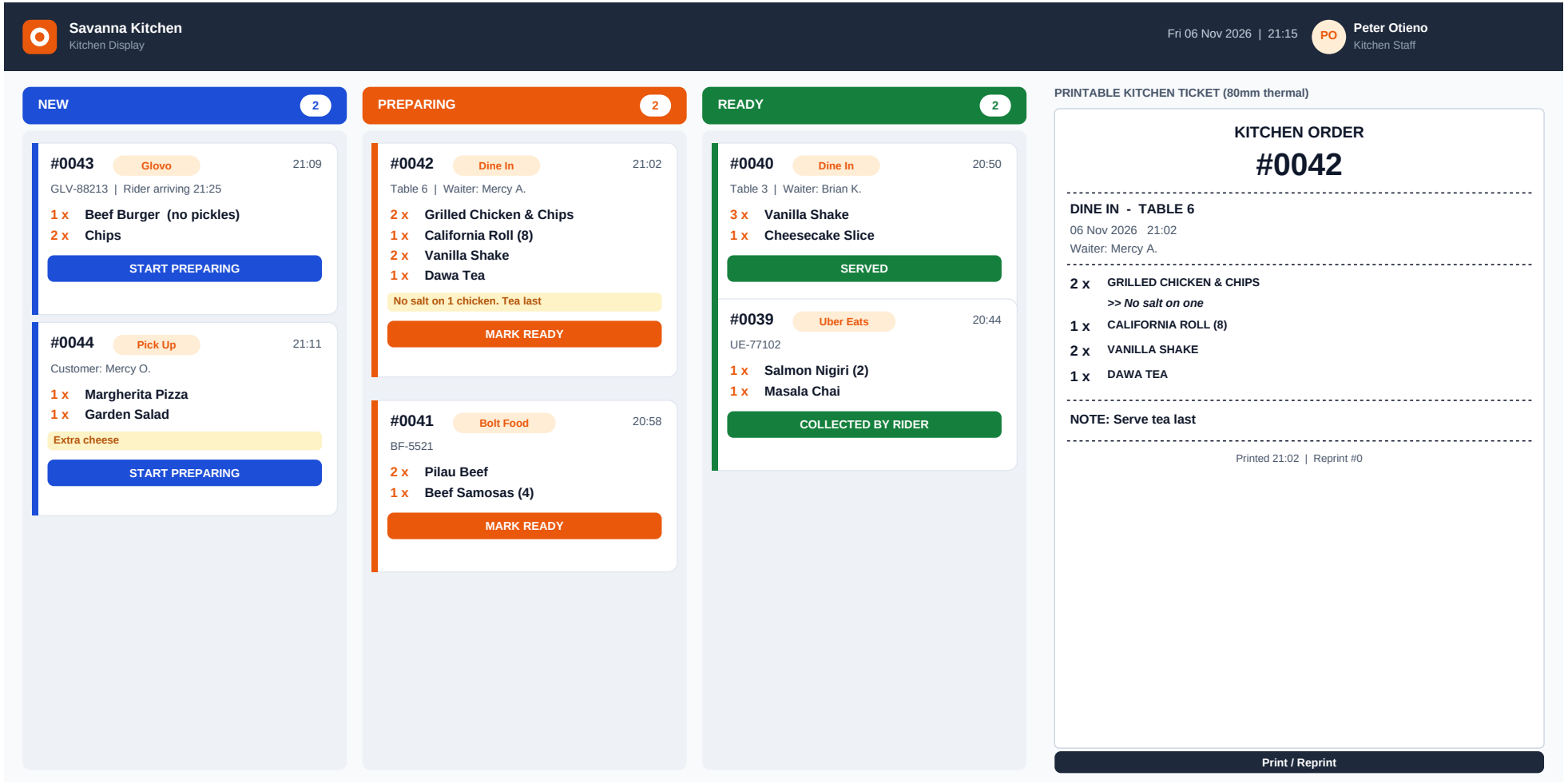


Purpose. Fast, large-button order entry for cashiers on a tablet.

What is on the screen. Order-type chips along the top, categories at left, search and item grid in the middle, cart at right. Tapping an item adds it; the orange badge shows the quantity in the cart. The cart shows the VAT portion, the total, payment method buttons and the Complete button. M-Pesa shows a code field; cash shows amount tendered and change.

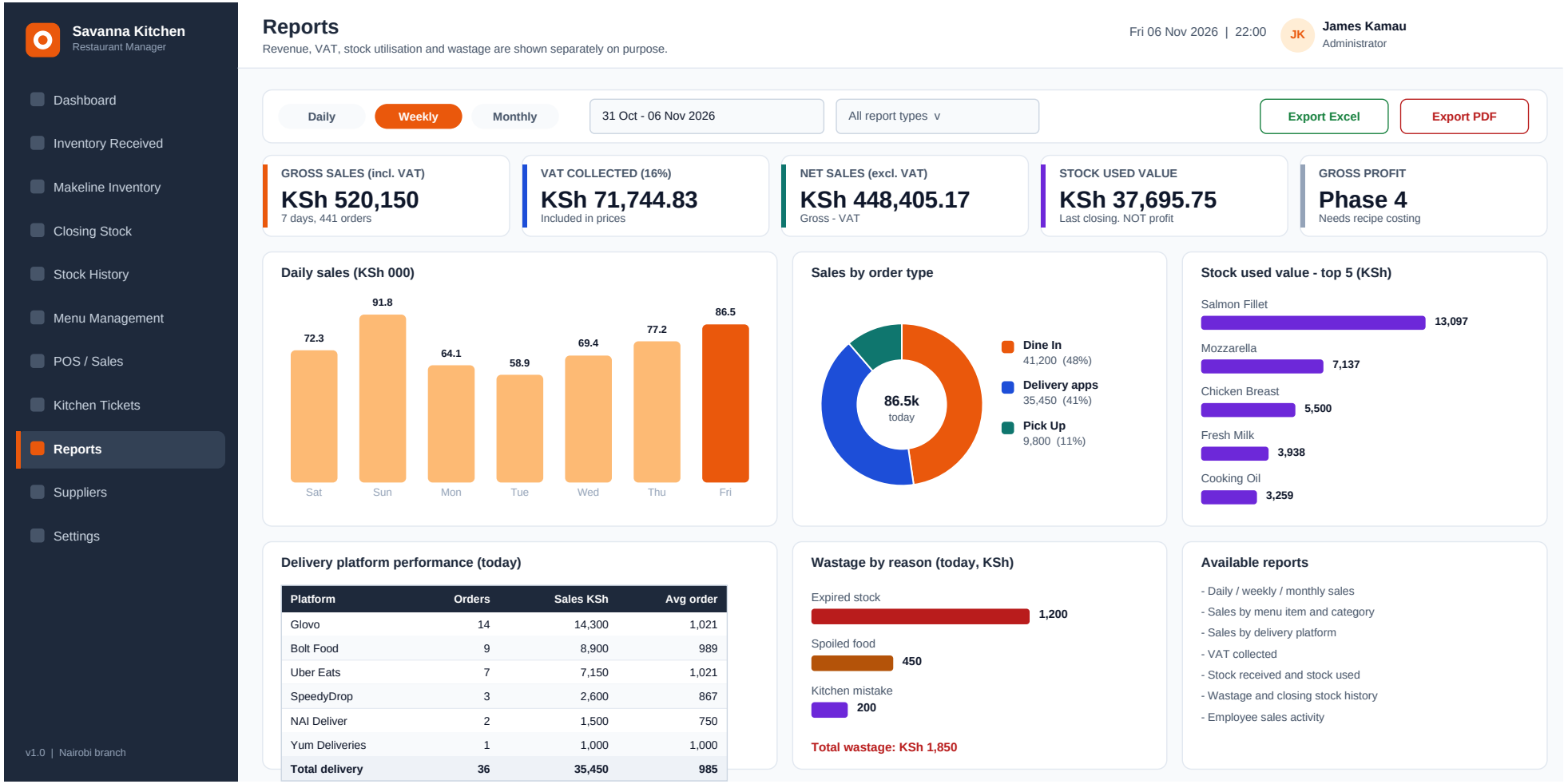
Developer notes. Menu items load from Menu Management. Unavailable items are greyed and cannot be added. The Complete button is disabled while saving. Idempotency key protects against double submit.

14.9 Screen 8 - Kitchen Tickets



Purpose. The kitchen sees what to cook and in what order, and updates progress with one tap.	**What is on the screen.** Three columns: New, Preparing, Ready. Each ticket shows number, order type, reference/table, time, items, special instructions. One large button moves a ticket to the next status. The right panel is the thermal-printer version of a ticket with a Print/Reprint button.	**Developer notes.** New tickets appear automatically with a short sound. Older tickets turn red after the configurable waiting time (default 15 minutes). Kitchen staff cannot see prices or payments.

14.10 Screen 9 - Reports

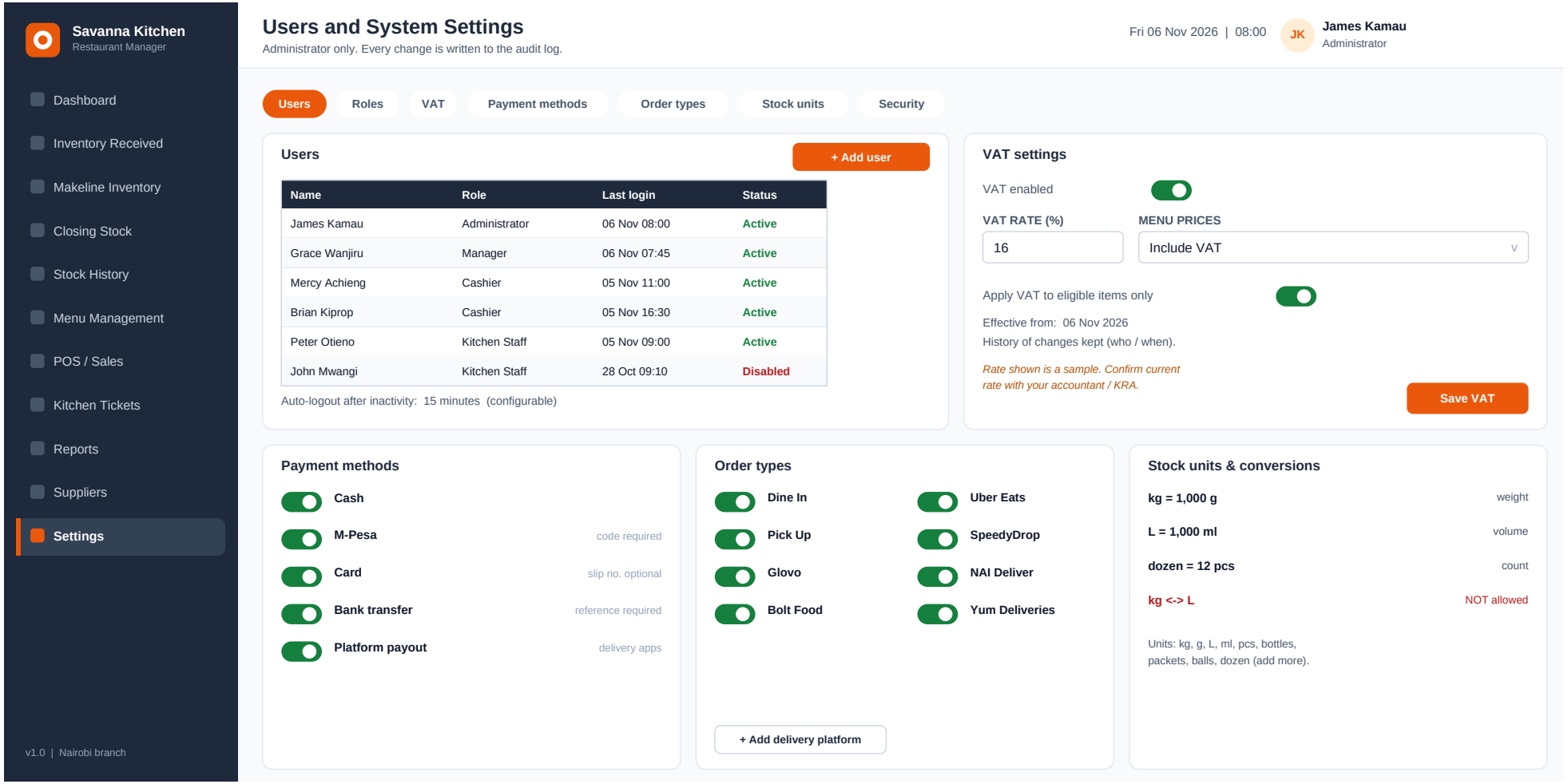


Purpose. Management reports with charts and exports.

What is on the screen. Choose Daily, Weekly or Monthly and a date range. KPI row separates gross sales, VAT, net sales, stock used value and gross profit. Charts show daily sales, order-type split, top stock used and wastage reasons. Delivery platform performance is a table. Export to Excel and PDF at the top right.

Developer notes. Gross profit shows "Phase 4" until recipes exist so nobody mistakes stock used for profit. All charts must be reproducible from the exported tables.

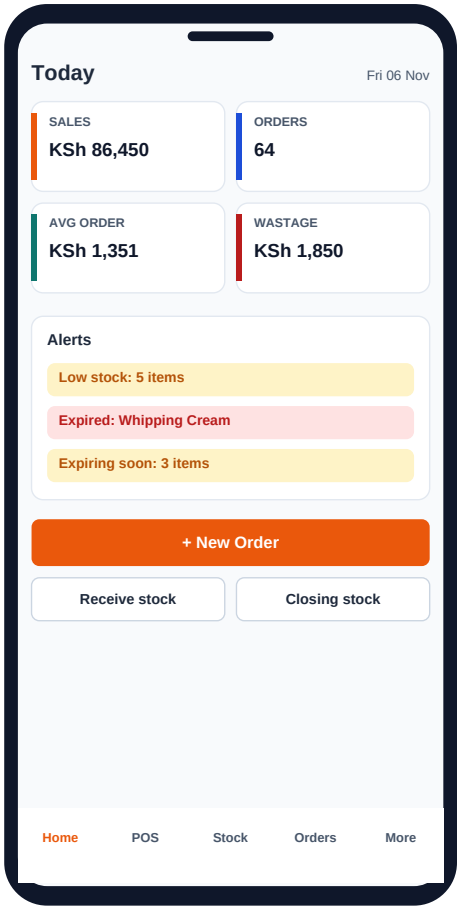
14.11 Screen 10 - Settings



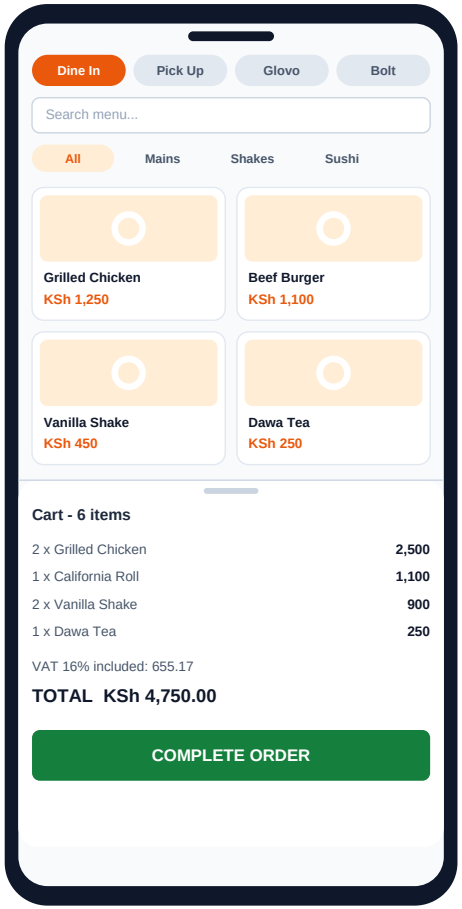
Purpose. Administrator control of users and system rules.	**What is on the screen.** Tabs for Users, Roles, VAT, Payment methods, Order types, Stock units and Security. Shown: user list with status, VAT settings with effective date, payment methods with reference rules, order types and delivery platforms with on/off toggles, and unit conversions. Incompatible conversions are shown as not allowed.	**Developer notes.** Every change writes to the audit log. Disabled users and inactive types stay in history. Administrator only.

14.12 Responsive mobile layouts

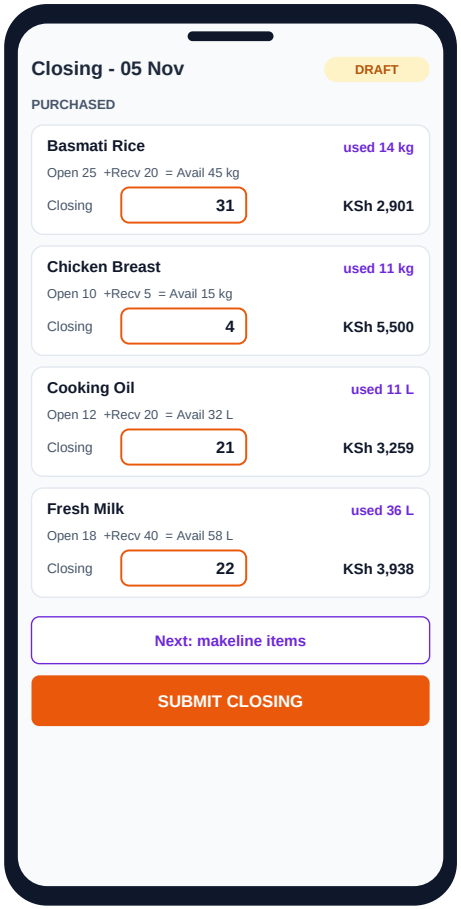
Dashboard (mobile)



POS (mobile)



Closing stock (mobile)



Purpose. The same system on a phone, for managers on the move and for waiters taking orders.	**What is on the screen.** Dashboard becomes stacked cards with alert chips. POS shows two item columns and a cart that slides up from the bottom with the total and Complete button always reachable. Closing stock becomes one card per product with the count box next to the used quantity and value.	**Developer notes.** Breakpoints: desktop above 1100 px, tablet 700-1100 px, phone below 700 px. Touch targets at least 44 px. No horizontal scrolling for any form.

15 Sample data

All figures in this document use the same fictional restaurant ("Savanna Kitchen") so that every example can be checked. The developer should load this data as a **demo database** for testing and training. Names, prices and suppliers are samples; the owner will supply real lists at setup.

15.1 Purchased products

Product	Category	Base unit	Min level	Usual supplier
Basmati Rice	Dry goods	kg	10	Mama Rosa Wholesalers
Chicken Breast	Meat & poultry	kg	5	Mama Rosa Wholesalers
Cooking Oil	Oils & fats	L	10	Mama Rosa Wholesalers
Fresh Milk	Dairy	L	15	Dairy Direct Ltd
Mozzarella	Dairy	kg	4	Dairy Direct Ltd
Salmon Fillet	Fish	kg	3	Ocean Catch Kenya
Tomatoes	Produce	kg	8	FreshFarm Produce Ltd
Nori Sheets	Dry goods	packets	10	Ocean Catch Kenya
Whipping Cream	Dairy	L	4	Dairy Direct Ltd

15.2 Makeline products

Makeline product	Unit	Description
Marinated Chicken	kg	Chicken marinated in ginger, garlic, lemon
Pizza Dough	balls	Rested dough balls for pizza
Sushi Rice (cooked)	kg	Seasoned sushi rice
Tomato Sauce	L	House tomato and basil sauce

15.3 Complete closing-stock sample (Thursday 05 November 2026)

Opening and received figures are the inputs; every other column is calculated. Use this table as the **acceptance test** for the closing calculation.

Product	Unit	Opening qty	Opening avg KSh	Received qty	Received KSh	Waste	Avail.	Closing	Used	Avg cost	Value used
Basmati Rice	kg	25	205.00	20	4,200.00	0	45	31	14	207.22	2,901.11
Chicken Breast	kg	10	500.00	5	2,500.00	0	15	4	11	500.00	5,500.00
Cooking Oil	L	12	290.00	20	6,000.00	0	32	21	11	296.25	3,258.75
Fresh Milk	L	18	108.00	40	4,400.00	0	58	22	36	109.38	3,937.66
Mozzarella	kg	4	1,050.00	6	6,780.00	0	10	3.5	6.5	1,098.00	7,137.00
Salmon Fillet	kg	3	2,350.00	5	12,000.00	0	8	2.5	5.5	2,381.25	13,096.88
Tomatoes	kg	8	110.00	15	1,800.00	1	22	6	16	116.52	1,864.35
Total					**37,680.00**						**37,695.75**

Closing value of stock carried to the next opening: KSh 27,546.74. Value of wastage: KSh 116.52 (Tomatoes, 1 kg, spoiled).

15.4 Makeline closing sample

Makeline product	Unit	Opening	Produced	Total available	Closing	Used
Marinated Chicken	kg	6	12	18	5	13
Pizza Dough	balls	20	40	60	14	46
Sushi Rice (cooked)	kg	8	15	23	6	17
Tomato Sauce	L	5	10	15	4	11

15.5 Menu sample

Category	Items and prices (KSh, VAT-inclusive)
Starters	Beef Samosas (4) 400 | Spring Rolls (6) 450 | Chicken Wings (8) 850
Main Meals	Grilled Chicken & Chips 1,250 | Beef Burger 1,100 | Margherita Pizza 1,300 | Pilau Beef 950
Milkshakes	Vanilla Shake 450 | Chocolate Shake 500 | Strawberry Shake 500
Sushi	Salmon Nigiri (2) 700 | California Roll (8) 1,100 | Salmon Avocado Roll 1,200
Teas	Dawa Tea 250 | Masala Chai 300 | Iced Tea 350
Mocktails	Virgin Mojito 600 | Passion Cooler 550
Desserts	Cheesecake Slice 550 | Ice Cream Scoop 300
Sides	Chips 400 | Garden Salad 450

15.6 Sample order and VAT

Order ORD-20261106-0042, Dine In, Table 6, cashier Mercy Achieng, paid by M-Pesa (code SJK4H2L9QX). VAT mode: inclusive, sample rate 16%, all items eligible.

Item	Qty	Price	Line total	VAT in line	Net
Grilled Chicken & Chips	2	1,250.00	2,500.00	344.83	2,155.17
California Roll (8)	1	1,100.00	1,100.00	151.72	948.28
Vanilla Shake	2	450.00	900.00	124.14	775.86
Dawa Tea	1	250.00	250.00	34.48	215.52
Order total			**4,750.00**	**655.17**	**4,094.83**

VAT is worked out line by line and rounded; the order VAT is the sum of the lines (check: 4,750.00 x 16 / 116 = 655.17).

15.7 Dashboard day (Friday 06 November 2026)

Measure	Sample value	Check
Sales / orders / average	KSh 86,450 / 64 / KSh 1,350.78	86,450 / 64
By order type	Dine In 41,200; Glovo 14,300; Pick Up 9,800; Bolt Food 8,900; Uber Eats 7,150; SpeedyDrop 2,600; NAI Deliver 1,500; Yum Deliveries 1,000	Adds to 86,450
By payment method	Cash 24,800; M-Pesa 17,500; Card 8,700; Platform payout 35,450	Adds to 86,450
Wastage today	Expired stock 1,200; Spoiled food 450; Kitchen mistake 200	Adds to 1,850

15.8 Suppliers and users

Supplier	Supplies	Payment terms (sample)
Mama Rosa Wholesalers	Rice, oil, chicken	7 days
Dairy Direct Ltd	Milk, cream, mozzarella	Cash on delivery
Ocean Catch Kenya	Salmon, nori	Cash on delivery
FreshFarm Produce Ltd	Vegetables, fruit	7 days

User	Role
James Kamau	Administrator / Owner
Grace Wanjiru	Manager
Mercy Achieng, Brian Kiprop	Cashier
Peter Otieno, John Mwangi	Kitchen Staff

16 Development phases

Build the core, make it reliable, and only then add more. Each phase must be **tested and accepted by the owner** before the next one starts. The durations below are indicative only; the developer must confirm estimates after reading this document.

Phase	Scope	Done when (exit criteria)	Indicative time
Phase 1 - Core operations	Login and basic users. Products, units, suppliers. Inventory Received. Makeline Inventory. Closing Stock (with weighted average). Stock History. Menu Management (items, categories, price history).	The manager can receive stock, record makeline, close the day and review history. Sample calculations match Section 15 to the cent. Backups running.	5-7 weeks
Phase 2 - Sales	POS. Order types and delivery platforms. VAT settings. Payments (cash, M-Pesa codes, card, bank, split). Receipts. Kitchen tickets and status. Sales history. Voids and refunds.	A full service can be run on the system. A cashier can complete an order in under 45 seconds. Printer works.	5-6 weeks
Phase 3 - Management	Dashboard. Reports with PDF/Excel export. Wastage screen. Low-stock and expiry alerts. Full roles and permissions. Audit log screen. Adjustments with approval.	Owner can see all dashboard figures and reports. Every important change shows in the audit log.	4-5 weeks
Phase 4 - Advanced	Recipe-based ingredient deduction. Supplier purchase analytics. Delivery platform integrations. Advanced profit and food-cost reporting. M-Pesa (Daraja) confirmation. KRA eTIMS integration (if required). Offline mode.	Scoped separately after Phases 1-3 are in daily use.	To be scoped

16.1 Go-live plan

1. **Data setup:** owner supplies product list, minimum levels, supplier list, menu with prices, users and roles. Developer loads them.
2. **Opening stock count:** full physical count with cost per unit (Section 5.5).
3. **Training:** one hour per role using the demo database; each employee completes the workflows in Appendix A for their role.
4. **Parallel run:** run the old method and the system side by side for 5-7 trading days; compare daily totals.
5. **Go-live:** after the owner signs the acceptance checklist (Section 18). Developer provides on-call support for the first two weeks.

OWNER DECISION REQUIRED

Decision: Which tablets, printers and Wi-Fi equipment will be used, and who buys them?

Recommended default: Owner buys one 10-inch tablet per cashier station, one 80 mm thermal printer per print point, and a backup mobile data router. The developer lists tested models in writing before purchase.

Owner to approve or change before development of this feature starts.

17 Testing requirements

The developer must test every item below and give the owner a short written test report before each phase is accepted. Tests use the sample data of Section 15 so the expected result is known in advance.

17.1 Calculation tests (must pass exactly)

ID	Test	Expected result
C1	Unit price: Cooking Oil, 20 L, KSh 6,000	KSh 300.00 per litre
C2	Chicken: open 10, receive 5 (KSh 2,500), count 4	Available 15; used 11; value KSh 5,500.00
C3	Cooking Oil: open 12 at 290, receive 20 for 6,000, count 21	Avg 296.25; used 11; value KSh 3,258.75; closing value 6,221.25
C4	Tomatoes with 1 kg wastage, count 6	Available 22; used 16; value KSh 1,864.35; wastage value 116.52
C5	Full sample closing (Section 15.3)	Total value used KSh 37,695.75; total received KSh 37,680.00
C6	500 g entered for a kg product	Stored as 0.5 kg; unit price per kg unchanged
C7	Enter kg for a litre-based product	Blocked with clear message
C8	VAT inclusive: KSh 4,750 at 16%	VAT 655.17; net 4,094.83
C9	VAT exclusive: net 1,000 at 16%	VAT 160.00; total 1,160.00
C10	Change VAT rate to 18% and reopen an old order	Old order unchanged (still shows 16%)

17.2 Business-rule tests

ID	Test	Expected result
B1	Closing count higher than Total available	Cannot submit; message names the product
B2	Submit closing twice (double click, or resend)	Only one closing created
B3	Edit a submitted closing as Manager without permission	Not possible; no edit option shown and server refuses
B4	Authorised correction of latest closing	New version created; old kept as superseded; reason stored; audit entry
B5	Submit closing, then open next day	Opening quantities equal the previous closing counts
B6	Try to delete a purchase, closing, order or audit entry	No delete option; direct database delete rejected for the app user
B7	Rename a product after a closing	Old closing still shows the old name and cost
B8	Change a menu price, view an order from yesterday	Old order shows old price
B9	Complete order with the same cart key twice	One order, same order number returned
B10	Two cashiers complete orders at the same moment	Different order numbers, no gaps from errors
B11	Void an order without a reason	Not accepted
B12	Makeline closing	No money value appears for makeline rows anywhere

17.3 Security, permission and reliability tests

- **Permissions:** log in as each of the four roles and confirm each action in the permission table (Section 11) is allowed or refused, including by typing the address directly.
- **Login:** wrong password lockout; automatic logout after the configured time; password never visible in logs or database.
- **Audit:** every action listed in Section 11.1 creates a log entry with user, time and before/after.
- **Backup and restore:** restore last night's backup on a clean machine; compare record counts and a sample closing.
- **Power or network loss** during order save and closing submit: no half-saved records.
- **Performance:** load 2 years of demo data (about 50,000 orders); searches under 1 second; POS tile tap response under 0.5 seconds.

- **Devices:** test on a tablet, a phone and a desktop in current Chrome and Safari; test thermal printing on the actual printers.

17.4 User acceptance testing

Real staff complete the workflows in Appendix A on the demo database. Record the time taken for each. The targets in Section 2 must be met. Any confusion or mistakes observed are logged and fixed before go-live.

18 Developer acceptance checklist

The owner signs each line when it has been demonstrated, not when it has been promised. Items marked **[Core]** are mandatory for the phase shown.

Done	Phase 1 - requirement
[ ]	[Core] Inventory received: all 12 receipt fields, unit price calculated automatically, multiple deliveries per product, search and filter, expiry highlighting, supplier returns and damaged deliveries.
[ ]	[Core] Purchase records cannot be deleted; corrections via adjustments with audit trail.
[ ]	[Core] Makeline inventory separate from purchased inventory; production history; wastage and adjustments; included in closing.
[ ]	[Core] Closing stock screen: alphabetical, two separate groups, all columns, automatic calculations match Section 15.3.
[ ]	[Core] Weighted average cost handles different purchase prices.
[ ]	[Core] Closing submit: saved, locked, next opening updated, time and user stored, history updated, no overwrite, authorised correction with reason.
[ ]	[Core] Stock history: date selection, compare days, search, export to Excel and PDF.
[ ]	[Core] Unit conversions work and incompatible units are blocked. No silent negative inventory.
[ ]	[Core] Menu management: all fields, price history, activate/deactivate, search and category views.

Done	Phase 2 - requirement
[ ]	[Core] POS: search, categories, items, cart, quantity controls, totals, discounts (authorised), VAT, payment, Complete.
[ ]	[Core] Eight order types; administrator can add or deactivate platforms.
[ ]	[Core] Order stores all fields in Section 8.2; unique order number; permanent storage; duplicate protection.
[ ]	[Core] VAT configurable (enable, rate, inclusive/exclusive, eligible items); no hard-coded rate.
[ ]	[Core] Payments: cash, M-Pesa, card, bank transfer, configurable others; paid/partial/unpaid/refunded; split payments.
[ ]	[Core] Kitchen tickets: contents, thermal print, status flow, order progress visible in POS.
[ ]	[Core] Sales history search by date, number, item, type, employee, payment method, status; voids and refunds with reason.

Done	Phase 3 - requirement
[ ]	[Core] Dashboard shows every item listed in Section 10.1.
[ ]	[Core] All reports in Section 10.2 with PDF and Excel export; stock used value labelled as not profit.
[ ]	[Core] Wastage screen with reason and employee; separate from consumption in reports.
[ ]	[Core] Low-stock and expiry alerts.
[ ]	[Core] Four roles with permission controls; audit log visible to Administrator.
[ ]	[Core] Automatic logout; password rules; HTTPS.

Done	Go-live - requirement
[ ]	[Core] Backups running and a restore test completed and recorded.
[ ]	[Core] Demo database, user guide (with Appendix A workflows) and training completed.
[ ]	[Core] Owner has approved every item in Appendix B.
[ ]	[Core] Source code, database design and administrator credentials handed to the owner.

Owner signature	Developer signature	Date

19 Future improvements

These are intentionally left out of the first versions. The structure in this blueprint is ready for them.

Improvement	What it adds	Prepared by
Recipe-based deduction	Selling a dish automatically deducts its ingredients and makeline items; shows theoretical vs actual usage (variance).	recipe_lines table; recipe_deducted flag on order lines
Food-cost and gross profit	Cost of each dish, food-cost %, gross profit per dish, category and day.	Recipes + weighted average cost
Supplier analytics	Price trends, best supplier per product, on-time delivery, spend per supplier.	purchase_items price history
Purchase orders	Create orders to suppliers from low-stock list, receive against the order.	suppliers, products, min levels
Delivery integrations	Orders from Glovo, Bolt Food, Uber Eats and others arrive in the POS automatically.	order_types, delivery_ref
M-Pesa and card integration	Automatic payment confirmation and reconciliation.	payments.reference
KRA eTIMS	Tax-compliant invoice submission from each order.	VAT per line, order data
Offline mode	Take orders when the internet is down and sync later.	idempotency keys on orders
Multiple branches	Several outlets, central menu, per-branch stock.	Add branch_id to transactional tables
Station routing	Tickets printed per kitchen station (sushi bar, grill, bar).	menu category to station mapping
Staff and tips	Service charge, tips, staff meals, cash-up and shift reports.	users, payments
Customer features	Loyalty, customer history, WhatsApp receipts (with consent).	customer table
Swahili interface	Full Swahili labels.	Language file from day one

A Appendix A - Step-by-step user workflows

Each workflow gives the exact clicks, entries and screen responses. They are also the user-acceptance test scripts (Section 17.4) and the basis of the staff training.

A. Receiving a supplier delivery

Who: Manager

Step	You click / enter	You see
1	Click **Inventory Received** in the sidebar.	Receipt form on the left; purchase history on the right.
2	Click the **Product** box and type "cook"; choose **Cooking Oil**.	Category fills in as "Oils & Fats". Unit suggests Litres.
3	Type **20** in Quantity. Check the unit is Litres.	Quantity accepted. Message appears if the unit is not compatible.
4	Type **6000** in Total purchase price.	Unit price shows **KSh 300.00 / L** at once (green, read-only).
5	Choose **Mama Rosa Wholesalers**; type invoice **INV-20418**.	Supplier saved; invoice must not be empty.
6	Check Date received (today) and enter the Expiry date **06 Aug 2027**.	Dates accepted. A past expiry date shows a warning.
7	Add a note if needed. Click **Save receipt**.	Green message "Receipt PUR-20261105-007 saved". The row appears at the top of the history; stock balance increases by 20 L.
8	For another product on the same invoice, click **Add another item**.	Form clears but keeps supplier and invoice.

Result: Purchase saved permanently with who entered it and when. Closing Stock shows the 20 L in the Received column.

B. Recording makeline production

Who: Kitchen lead or Manager

Step	You click / enter	You see
1	Click **Makeline Inventory**.	Production form and makeline stock list.
2	Choose **Marinated Chicken** in Makeline product.	Unit shows kg (read-only).
3	Type **12** in Quantity produced.	Accepted.
4	Leave date/time as now. Choose your name in Person responsible.	Name defaults to the logged-in user.
5	Type a note ("ginger, garlic, lemon") and click **Save production**.	Row added to Production history; Available now rises by 12 kg.
6	If some was spoilt, click **Record wastage**, choose item, enter quantity and reason.	Wastage entry saved with reason; balance falls.

Result: Makeline stock updated and included in the next closing count. No money value is created.

C. Completing daily closing stock

Who: Manager

Step	You click / enter	You see
1	After service, click **Closing Stock**.	Screen opens with all purchased products (A) and makeline products (B), alphabetical. Opening, Received, Waste and Total available are already filled.
2	Physically count the first product. Click its orange Closing box and type the count (e.g. Basmati Rice **31**).	Used (14) and Value used (KSh 2,901.11) appear immediately.
3	Press Tab to move to the next row and repeat for all purchased products.	Product names go amber if at or below minimum stock. Summary panel updates.
4	Scroll to Makeline (B) and enter each count (e.g. Pizza Dough **14**).	Used quantity appears. No money value shown.
5	If a count is higher than Total available the box turns red.	Message: "Closing cannot be more than available (45 kg)." Fix by entering the missing receipt or an adjustment first.
6	Check the "Before you submit" list. Click **Save draft** to stop and continue later.	Draft saved; nothing is locked.
7	Click **SUBMIT CLOSING STOCK**.	Confirmation: "Submit closing for 05 Nov 2026? It will be locked." Click Confirm.
8	Wait for the green message.	"Closing CLS-20261105-01 submitted by Grace Wanjiru at 23:12." Screen is read-only. Next opening stock is set.

Result: Permanent locked snapshot in Stock History. Next day opens with these counts as Opening.

D. Reviewing historical stock

Who: Manager or Owner

Step	You click / enter	You see
1	Click **Stock History**.	List of past closings on the left; latest selected.
2	Click the Date box and choose **05 Nov 2026**.	Tiles and table show that day's closing exactly as saved.
3	Choose **Compare with: 04 Nov**.	Extra columns show the previous day's value used and % change; big changes in red.
4	Type "salmon" in Search product.	Table shows only Salmon Fillet.
5	Click **Export Excel** or **Export PDF**.	File downloads with the same figures, restaurant name, date and who exported.

Result: Owner can see opening, received, used, wasted and value used for any day, and compare days.

E. Adding a menu item

Who: Manager or Administrator

Step	You click / enter	You see
1	Click **Menu Management**.	Item form on the left, list on the right.
2	Type **Salmon Avocado Roll** as Item name.	Accepted.
3	Choose **Sushi** as Category.	Category set. Use "+ New category" for a new one.
4	Type **1200** as Selling price.	Shown as 1,200.00.
5	Leave VAT as Standard. Optionally add description, code SUS-004 and an image.	Image preview appears.
6	Make sure Available for sale is on. Click **Save menu item**.	Item appears in the list and, at once, in the POS under Sushi. A first price-history row is stored.

Result: The item can be sold immediately. A later price change adds a history row.

F. Processing a dine-in order

Who: Cashier

Step	You click / enter	You see
1	On the POS tap **Dine In**.	Chip turns orange; table box is required.
2	Type table number **6**.	Accepted.
3	Tap **Grilled Chicken & Chips** twice, then **California Roll**, **Vanilla Shake** twice, **Dawa Tea**.	Items appear in the cart with orange badges on the tiles.
4	Tap the line "Grilled Chicken & Chips" and add note "no salt on one".	Note appears under the line in amber.
5	Check totals.	Total KSh 4,750.00 with "VAT 16% included 655.17".
6	Tap **Send to kitchen** (if the guests pay later).	Kitchen ticket #0042 created; order shows status New.
7	When guests pay, tap **M-Pesa**, type the code **SJK4H2L9QX**.	Payment row added; status changes to PAID.
8	Tap **Complete order + receipt**.	Order number ORD-20261106-0042 shown; receipt prints. Order moves to sales history.

Result: Order stored permanently; kitchen ticket created; receipt printed.

G. Processing a delivery platform order

Who: Cashier

Step	You click / enter	You see
1	Tap the platform chip, e.g. **Glovo**.	Delivery reference box appears and is required.
2	Type the platform order number **GLV-88213**.	Accepted.
3	Tap **Beef Burger** and **Chips** x 2; add note "no pickles".	Cart updates; VAT and total calculated.
4	Choose payment method **Platform payout** (or the agreed method).	Order shown as Unpaid / platform until reconciled, or Paid if the platform paid at order time (owner decision).
5	Tap **Complete order**.	Order number issued; kitchen ticket #0043 prints with "GLOVO GLV-88213".
6	When the rider arrives, open the Ready list and tap **Collected by rider**.	Status Collected; ticket leaves the kitchen board.

Result: Delivery orders appear in Sales by delivery platform.

H. Printing a kitchen ticket

Who: Cashier or Kitchen Staff

Step	You click / enter	You see
1	After an order is confirmed, look at **Kitchen Tickets**.	Ticket shown in the New column with number, type, items, notes.
2	If auto-print is on, the ticket prints. If not, click the ticket then **Print / Reprint**.	Preview of the 80 mm ticket appears at right; printer outputs it.
3	Kitchen taps **Start preparing**.	Ticket moves to Preparing; POS shows Preparing.
4	Kitchen taps **Mark ready**.	Ticket moves to Ready; POS highlights it.
5	Waiter taps **Served**.	Ticket leaves the board.

Result: Reprints are counted on the ticket ("Reprint #1") so duplicates are known.

I. Reviewing daily sales

Who: Manager or Owner

Step	You click / enter	You see
1	Click **Dashboard**.	Today's sales, orders, average order, sales by type and payment, best sellers.
2	Click **Reports** and choose Daily; pick the date.	KPI row and charts for that day.
3	Check Gross sales, VAT collected and Net sales.	VAT and net shown separately. Stock used value is shown and labelled "not profit".
4	Click **Delivery platform performance** row to see platform orders.	List of orders for that platform.
5	Click **Export PDF** or **Export Excel**.	File with the same figures downloads.

Result: Owner sees sales, VAT, orders and platform performance for any day.

J. Correcting a stock error

Who: Manager (with permission)

Step	You click / enter	You see
1	Example: 20 L of oil was entered as 2 L. Click **Inventory Received** and open the receipt.	Receipt is read-only (padlock icon). Button **Create adjustment** is shown.
2	Click **Create adjustment**; choose direction **IN**; quantity **18** L.	System shows the cost to be used (average cost) and the new on-hand quantity.
3	Type the reason: "Receipt INV-20418 entered as 2 L, delivered 20 L." Click **Save adjustment**.	If the manager lacks authority, status is "Awaiting approval".
4	Administrator opens **Approvals** and clicks **Approve**.	Adjustment becomes effective; audit log shows who requested, who approved, when, and why.
5	Wrong closing already submitted? Open the latest closing in Stock History and click **Request authorised correction**.	Correction form requires a reason; a new version is created and the old one marked superseded.

Result: Original records untouched. Corrections are new records with reason and approver.

B Appendix B - Owner decisions register

Where information was missing, this blueprint chose a recommended default. The owner must approve or change each item **before the related feature is built**. Tick "Approved" when accepted, or write the change in the last column.

#	Decision	Recommended default	Section	Approved
1	KRA eTIMS requirement and receipt format	Record KRA PIN; leave eTIMS integration for Phase 4 unless the accountant says it is required now.	1.2	[ ]
2	VAT registration, rate, and prices inclusive or exclusive	Registered with sample 16% rate; menu prices VAT-inclusive; rate editable in Settings.	7.2 / 8.5	[ ]
3	Costing method for stock used	Weighted average cost.	4.3	[ ]
4	Treatment of wastage in reports	Separate column; "Total stock cost out" line shows used + wasted.	4.1	[ ]
5	Expiry warning period and FIFO assumption	5 days; first in, first out.	4.6	[ ]
6	Closings per day and cut-off	One closing per day; period runs from last closing to this one; shift closings can be switched on later.	4.7	[ ]
7	Who may correct a closing and how far back	Manager with permission: latest closing only. Older: dated adjustment, Administrator approves.	5.4	[ ]
8	Source of opening stock cost at go-live	Cost per unit from the latest supplier invoice for each product.	5.5	[ ]
9	Minimum stock level for each product	Owner supplies list; default 20% of weekly use until then.	4.6	[ ]
10	Discount authority	Only Manager/Admin; reason required; percentage cap set by owner (suggested 20%).	8.6	[ ]
11	Delivery platform sales and commissions	Sales recorded at menu price; platform commission not tracked in version 1.	8.1	[ ]
12	Payment status of delivery orders	Created as "Platform payout" and marked Paid when reconciled.	A-G	[ ]
13	Service charge, tips	Not included in version 1.	19	[ ]
14	Order number format	ORD-YYYYMMDD-NNNN, resets daily.	8.2	[ ]
15	Receipt content (name, address, PIN, footer text)	Owner supplies text; sample used here.	9	[ ]
16	Kitchen tickets, printers and drinks-only orders	One kitchen printer; drinks-only orders do not print tickets.	9	[ ]
17	Offline mode	Not in version 1; keep backup connection.	8.4	[ ]
18	Hosting location and backup responsibility	Cloud hosting with encrypted nightly backups; owner keeps admin credentials.	12.3	[ ]
19	Inactivity logout time	15 minutes (POS: 30 minutes with PIN resume).	11.1	[ ]
20	Customer phone numbers: who may view	Manager and Admin only; not in routine exports.	1.2	[ ]
21	Initial lists (products, menu, suppliers, users)	Owner supplies spreadsheet; developer loads it.	16.1	[ ]